**Prague University of Economics**
Faculty of Informatics and Statistics
Department of Statistics and Probability

# Bachelor's Thesis

# Comparison Between Frequentist and Bayesian Logistic Regression on the Example of Real Data

Mikhail Fedorov

Supervisor: Ing. Tomáš Karel, Ph.D.

Study Field: Mathematical Methods in Economics
Specialisation: Data Analyses and Modeling

Prague 2024

## Acknowledgements

I would like to express my deepest gratitude to my supervisor, Ing. Tomáš Karel, Ph.D., for his valuable advice, willingness to help, and to my family for support throughout the process of writing this thesis.

# Abstract

This study compares frequentist and Bayesian approaches to logistic regression using real-world data, with a particular focus on the impact of informative and non-informative priors in Bayesian analysis. The research is motivated by advances in computational capabilities that have made Bayesian methods increasingly accessible in machine learning applications.

The methodology involved splitting the dataset into two parts: a primary subset (20%) for model evaluation and a larger auxiliary subset (80%) simulating prior knowledge. Three approaches were compared: classical logistic regression, Bayesian logistic regression with non-informative priors, and Bayesian logistic regression with informative priors. Model performance was evaluated using multiple metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC score, along with an analysis of 95% confidence intervals (CI) as well as credible intervals .

Results demonstrate that while classical and non-informative prior Bayesian approaches showed similar performance, the Bayesian model with informative priors achieved superior results with an accuracy of 0.83 (compared to 0.78), F1-score of 0.85 (versus 0.81), and notably higher ROC-AUC of 0.93 (versus 0.81). The most significant improvement was observed in parameter uncertainty estimation, where the credible interval length undergone an average reduction of 15.1% compared to the classical approach, with improvements ranging from 10.3% to 28.2% across different variables.

# Keywords

Bayesian statistics, logistic regression, informative priors, machine learning, confidence intervals

# Contents

# Introduction

Logistic regression remains one of the most widely used machine learning methods for binary classification tasks. While rooted in classical probability theory, which has been a cornerstone of mathematical study for centuries, modern applications of this method have evolved to cover both frequentist and Bayesian paradigms.

The Bayesian approach to statistical inference has gained significant popularity, offering distinct advantages over traditional methods. Its primary strength lies in the ability to incorporate prior knowledge into the modeling process, making it particularly valuable in scenarios with limited data or significant uncertainty. Furthermore, the Bayesian interpretation of probability as a degree of belief, rather than the frequentist view of probability as long-run frequency, provides a more intuitive framework for applied tasks such as risk prediction and decision-making in real-world contexts.

However, the implementation of Bayesian methods, including Bayesian logistic regression, has historically been constrained by computational challenges. The complexity of calculating posterior distributions through direct integration of Bayes' theorem often makes analytical solutions intractable. This limitation has necessitated the development of sophisticated numerical methods for posterior approximation.

Recent advances in computational capabilities have revolutionized the practical application of Bayesian methods. Modern sampling techniques, including Markov Chain Monte Carlo (MCMC) methods such as the Metropolis-Hastings algorithm and Gibbs sampling, along with more advanced approaches like Hamiltonian Monte Carlo (HMC) and its No-U-Turn Sampler (NUTS) variant, have made complex Bayesian models computationally feasible. These developments have significantly expanded the scope of Bayesian statistics in addressing previously intractable problems.

This research aims to conduct a comprehensive comparison between Bayesian and classical logistic regression approaches, with both theoretical analysis and practical application to medical data. The medical domain serves as an ideal context for this comparison, as improvements in predictive accuracy can directly impact patient outcomes. Our study specifically focuses on cardiovascular disease prediction, where the stakes of accurate classification are particularly high.

The theoretical component of this work examines the fundamental differences between Bayesian and frequentist approaches, analyzing their respective strengths and limitations. The practical component demonstrates these differences through empirical analysis, with particular attention to:

- The incorporation of informative priors derived from expert knowledge and previous studies
- Comparative analysis of model performance using metrics such as ROC curves
- Assessment of parameter uncertainty through posterior analysis

By combining theoretical strictness with practical application, this research contributes to both the academic understanding of logistic regression methodologies and the development of more effective tools for medical diagnosis. The findings aim to demonstrate how Bayesian methods can enhance predictive modeling in healthcare, potentially leading to improved patient treatment outcomes through more accurate and interpretable predictions.

# 1 Theory of Probability

Probability theory is a part of mathematics designed to formalize the concepts of randomness and precisely handle the notion of probability as a quantity that measures this randomness. Each event is assigned a number from 0 to 1, where 0 means the event will certainly not occur, and 1 means the event will certainly occur. This scale exactly corresponds to the scale from 0 to 100 percent, and makes it much easier and intuitive for people to understand.

## 1.1 Axioms of Probability Theory

To establish a clear foundation for the axioms of probability theory, we will first outline several key definitions.(Bolstad,2007).

- **Random Experiment**: An experiment whose outcome cannot be predicted precisely in advance, although all possible outcomes are known. It is possible to repeat this experiment under the same conditions and get different results.
- **Outcome**: The result of one single trial of the random experiment.
- **Sample Space**: An arbitrary set, representing the set of outcomes of one single trial of a random experiment, denoted as $\Omega$.
- **Random Event**: Sets containing outcomes of random experiments which, upon conducting the random experiment, will either occur or not occur, and whose occurrence depends on chance.
- **Elementary Event**: The most basic possible result of a random experiment, representing one and only one of the possible outcomes.

## 1.2 Event Operations

Given two events $A$ and $B$, the following operations can be defined:

- **Union of Two Events**: The set of outcomes in either $A$ or $B$ (inclusive or), denoted $A \cup B$.
- **Intersection of Two Events**: The situation where two events $A$ and $B$ happen simultaneously, denoted $A \cap B$.
- **Complement of an Event**: The set of outcomes not in $A$, denoted as $A^{\complement}$ or $\overline{A}$.

## 1.3 Axioms of Probability

The three primary axioms of probability theory defined by (Kolmogorov,1933) :

1. **Non-negativity**: For any event $A$, the probability $P(A)$ is non-negative:
$$P(A) \geq 0.$$
2. **Normalization**: The probability of the entire sample space $\Omega$ is 1:
$$P(\Omega) = 1.$$
3. **Additivity**: For any sequence of mutually exclusive (disjoint) events $A_1, A_2, A_3, \ldots$:
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

## 1.4 Joint Probability and Independent Events

Following definitions were taken from (Bolstad,2007)
The *joint probability* of two or more events is the probability of the intersection of those events $P(A \cap B)$.
If they are *independent*, then $P(A \cap B) = P(A) \cdot P(B)$.
Marginal probability refers to the probability of an event occurring regardless of the outcome of another variable, and by using axioms of probability could be found:

- $A = (A \cap B) \cup (A \cap B^c)$
- $P(A) = P(A \cap B) + P(A \cap B^c)$

## 1.5 Conditional Probability

For the general definition, take events A and B, and assume that P(B) > 0. The conditional probability of A given B equals:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \tag{1.1}$$

where: - $P(A|B)$ is the probability of event $A$ occurring given that $B$ has occurred. - $P(A \cap B)$ is the probability of both events $A$ and $B$ occurring. - $P(B)$ is the probability that event $B$ occurs.
In case of conditional probability of *independent* events:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A). \tag{1.2}$$

Knowledge about A does not affect the probability of B occurring (Bolstad, 2007).

## 1.6 Classical Definition of Probability:

In the case of a finite number of possible outcomes of a random experiment, that is, in the case of a finite sample space of elementary events $E = \{E_1, E_2, E_3, \ldots, E_n\}$, where each elementary event $E_i$ has the same probability of occurring ($P(E_1) = P(E_2) = \ldots = P(E_n)$), we calculate the probability $P(A)$ of a random event $A$, which is a subset of the sample space $E$, using the formula:

$$P(A) = \frac{N_A}{N}. \tag{1.1}$$

where $N$ is the total number of all possible outcomes of the random experiment, that is, the total number of elementary events $E$ and $N_A$ is the number of results of the random experiment favorable to the event $A$, that is, the number of elementary events that make up event $A$ (Bílková, 2009).
The classical definition of probability is one of the earliest and simplest forms of defining probability. It serves as a fundamental concept to understand it. However, it is constrained by its assumptions of a finite number of outcomes and equal likelihood.

## 1.7 Probability as a Relative Frequency:

The difficulties encountered with the classical definition of probability mentioned above led to the interpretation of the frequency. Formally, the probability of an event A would be the limit of the relative frequency of occurrence of that event as the number of observations grows large.

$$P(A) = \lim_{N \to \infty} \frac{N_A}{N}. \tag{1.2}$$

This notion of probability is appealing because it seems objective and ties our work to the observation of physical events. We interpret probability as the proportion of times that such an event is expected to occur in the long run. The only drawback is that we are not capable of performing experiments unlimitedly (Bílková, 2009).

### 1.8 Subjective Definition of Probability.

Bayesian statistics is based on the concept of a subjective approach to probability. In this concept, probability is viewed as a degree of belief or confidence that a specific situation or phenomenon will actually occur. This idea was formulated by Jacob Bernoulli in his book "The Art of Prediction" (Ars Conjectandi, 1713). Different individuals can thus have different opinions, and therefore different subjective probabilities, about whether an event will or will not occur. However, the subjective perception of probability in terms of degree of belief does not in any way mean rejection of probability axioms in terms of Kolmogorov's definition of probability, or any other disregard for probability properties or non-compliance with rules for probability operations, from which Bayes' formula itself is derived.

The subjective view of probability has certain advantages compared to the classical approach. One advantage is the possibility of changing one's opinion (subjective probability) after obtaining new information, which is crucial for the Bayesian way of thinking (updating rule). Another advantage is that subjective probability can be quantified even when the person concerned has a certain opinion (knowledge or belief) about the given issue. Thanks to this fact, it is possible to assign probability even to otherwise unrepeatable phenomena, where other approaches to the concept of probability fail by their very definitions (e.g., frequency and classical definitions of probability). As Hebák (2012) further points out, both classical and statistical (frequency) definitions of probability contain tautology and are essentially circular definitions. The classical definition of probability has a built-in prior assumption of equal probability for individual elementary events, while the frequency definition assumes independence of individual trials. Both these assumptions, however, cannot be verified without knowledge of the concept of probability.(Karel, 2017)

## 2 Bayesian statistics

### 2.1 Bayes Theorem

The Reverend Thomas Bayes first discovered the theorem that now bears his name, originally presenting it in an essay on "Solving a Problem in the Doctrine of Chances." Posthumously published in the Philosophical Transactions of the Royal Society in 1763 by his friend Richard Price, Bayes' work demonstrated how inverse probability could be used to calculate the probabilities of antecedent events based on the occurrences of consequent events. While his methods were initially adopted by Laplace and other scientists in the 19th century, they had largely fallen from favor by the early 20th century.

In probability theory, conditional probability becomes crucial when we need to adjust our understanding based on additional information (Bolstad, 2007). This is precisely where Bayes' theorem becomes invaluable.

Bayes' theorem is fundamentally rooted in conditional probability, providing a powerful framework for probabilistic reasoning. Let $B_1, B_2, \ldots, B_n$ represent a set of possible hypotheses under which event $A$ might occur. The theorem allows us to compute the probabilities of each hypothesis $B_i$ given that event $A$ has occurred.

The core of Bayes' theorem can be expressed through several equivalent formulations: For a specific hypothesis $B$, the basic form is:

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}. \tag{2.1}$$

A more detailed version incorporating is:

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|\overline{B}) \cdot P(\overline{B})}. \tag{2.2}$$

For a comprehensive set of mutually exclusive events, the generalized form becomes:

$$P(B_i \mid A) = \frac{P(A \mid B_i)\, P(B_i)}{\sum_{i=1}^{n} P(A \mid B_i)\, P(B_i)}. \tag{2.3}$$

Let's break down the key components of these formulations:

- $P(B|A)$ represents the posterior probability—the probability of event $B$ given that event $A$ has occurred.
- $P(A|B)$ is the likelihood—the probability of event $A$ occurring given that event $B$ is true.
- $P(B)$ denotes the prior probability—the initial probability of event $B$ before obtaining new evidence.
- $P(A|\overline{B})$ is the probability of event $A$ occurring when event $B$ is false.
- $P(\overline{B})$ represents the probability that hypothesis $B$ is not true.

At its core, Bayes' theorem provides a systematic method to update our prior beliefs about the probabilities of hypotheses based on new empirical evidence. It serves as a fundamental tool in Bayesian statistics, enabling more nuanced inference and decision-making processes (Bolstad, 2007).

Following picture schematically depicts this process:

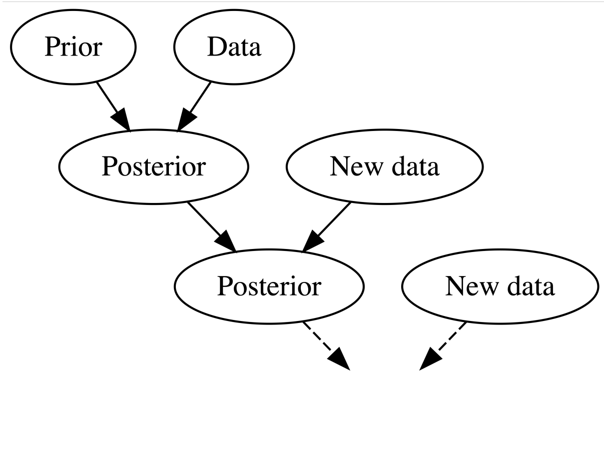


Figure 2.1: A Bayesian knowledge-building diagram.

Source: Johnson, Ott & Dogucu , 2022

**Bayes' Theorem for Continuous Variables** While the discrete form of Bayes' theorem is widely used, the theorem can be extended to continuous random variables through probability density functions. For a continuous random variable $X$ and parameter $\theta$, the continuous version of Bayes' theorem is expressed as:

$$p(\theta|X) = \frac{p(X|\theta) \cdot p(\theta)}{p(X)}, \tag{2.4}$$

where:

- $p(\theta|X)$ is the posterior probability density.
- $p(X|\theta)$ is the likelihood function.
- $p(\theta)$ is the prior probability density.
- $p(X)$ is the marginal likelihood (evidence).

The marginal likelihood $p(X)$ can be calculated by integrating over all possible parameter values:

$$p(X) = \int p(X|\theta) \cdot p(\theta)\, d\theta. \tag{2.5}$$

## 2.2 Bayesian Inference

Bayesian statistics represents a sophisticated and intuitive approach to statistical analysis that has transformed our understanding of data interpretation. Unlike traditional frequentist methods, Bayesian statistics enables researchers to make direct probabilistic statements about parameters of interest and incorporate prior knowledge into their analyses. At its core, the Bayesian framework elegantly combines two essential sources of information: prior knowledge (expressed as a prior distribution) and observed data (captured in the likelihood function), resulting in a posterior distribution that represents our updated understanding.

What sets Bayesian statistics apart is its treatment of parameters as random variables with their own probability distributions, rather than fixed but unknown quantities. This fundamental difference allows for more natural and intuitive interpretation of uncertainty in statistical inference. Through the Bayesian lens, we can continuously update our beliefs as new evidence emerges, making it particularly valuable in fields requiring dynamic decision-making under uncertainty.

The practical advantages of this approach include:

- Direct probability statements about parameters of interest.
- Natural incorporation of prior knowledge and expertise.
- Intuitive framework for updating beliefs with new data.
- Robust handling of uncertainty in statistical inference.
- Flexible modeling of complex hierarchical structures.

This probabilistic reasoning framework has proven particularly powerful in modern applications, from machine learning to scientific research, offering a more comprehensive and nuanced approach to statistical inference (Bolstad,2007).

## 2.3 Introduction to Bayesian Inference

One of the most important aspects of modeling using Bayes theorem is Bayesian inference. Its importance is conditioned, first of all, by the possibility to update knowledge when new data are obtained. Unlike frequency statistics, Bayesian statistics allows to take into account a prior ideas about model parameters, which can significantly increase the accuracy of predictions ( Johnson et al. , 2022 ).

Bayesian inference is based on three key components (Bolstad, 2007):

- $g(x_i)$ - prior distribution.
- $f(x_i|y_i)$ - posterior distribution.
- $f(y_i|x_i)$ - likelihood.

According to Bayes' theorem, the posterior distribution is obtained by adjusting the prior distribution for the probability of the observed data. In functional form, we can write as:

$$f(x_i, y_i) = g(x_i) \times f(y_i|x_i). \tag{2.6}$$

Bayesian statistics and probabilistic models often use this formula:

$$f(x_i, y_i) = \frac{g(x_i) \times f(y_i|x_i)}{\sum_{i=1}^{n_i} g(x_i) \times f(y_i|x_i)}. \tag{2.7}$$

This formula is especially useful when our goal is to find the marginal distribution of $y_i$ given $x_i$. We also need to normalize the result so that the sum of probabilities across all $y_i$ equals one. In this case, the denominator represents the marginalization over $x_i$, meaning it calculates the total probability of the observed event $y_i$ without conditioning on a specific $x_i$.

For continuous random variables, Bayes' theorem takes the following form:

$$f(\theta|x) = \frac{f(x|\theta)g(\theta)}{\int_{-\infty}^{\infty} f(x|\theta)g(\theta)d\theta}, \tag{2.8}$$

where:

- $f(\theta|x)$ is the posterior distribution of parameter $\theta$ given the data $x$,
- $f(x|\theta)$ is the likelihood function,
- $g(\theta)$ is the prior distribution,
- The denominator $\int_{-\infty}^{\infty} f(x|\theta)g(\theta)d\theta$ is the marginal likelihood or evidence.

This continuous form is particularly important in real-world applications where parameters can take any value within a continuous range, such as in physical measurements, economic indicators, or biological parameters (Bolstad, 2007).

## 2.4 Analytical Intractability Challenge

A critical challenge in Bayesian inference for continuous variables emerges in the calculation of the marginal likelihood $p(X)$ also known as normalizing constant:

$$p(X) = \int p(X|\theta) \cdot p(\theta)\, d\theta. \tag{2.9}$$

This integral becomes analytically intractable for most complex models due to several key reasons:

- **High-Dimensional Integrals:** As the number of parameters increases, the computational complexity grows exponentially.
- **Non-Conjugate Priors:** When prior and likelihood functions do not belong to the same probability distribution family, analytical integration becomes impossible.
- **Complex Likelihood Functions:** Many real-world models have likelihood functions that prevent closed-form solutions.

## 2.5 Computational Solutions

To address these challenges, researchers have developed several computational approaches:

- **Markov Chain Monte Carlo (MCMC):** Numerical sampling techniques like Metropolis-Hastings and Gibbs sampling.
- **Variational Inference:** Approximation methods that transform the integration problem into an optimization problem.
- **Approximate Bayesian Computation (ABC):** Simulation-based techniques for models with intractable likelihoods.

In practice, these computational methods allow researchers to perform Bayesian inference in scenarios where analytical solutions are impossible(Caflisch, 1998).

## 2.6 Markov Chain Monte Carlo

The development of Markov Chain Monte Carlo (MCMC) methods revolutionized Bayesian statistics by overcoming the computational challenges that previously limited its practical application. These methods provide a systematic approach to approximate complex posterior distributions, which are often analytically intractable, particularly in high-dimensional parameter spaces. Algorithms such as the Hamiltonian Monte Carlo (HMC) and the Gibbs sampler have become cornerstone techniques within the Bayesian framework. HMC leverages the principles of physics to efficiently explore posterior distributions, making it well-suited for models with a large number of parameters. On the other hand, the Gibbs sampler simplifies the sampling process by breaking it into conditional distributions, which are often easier to handle. Together, these tools have empowered researchers to tackle problems in fields ranging from genetics and epidemiology to machine learning and finance, enabling Bayesian methods to flourish in real-world, data-intensive scenarios(Neal , 2011).

### 2.6.1 Markov Chain

Markov Chain is a stochastic model which describes a sequence of events in which the probability of each subsequent state depends only on the current state and not on the preceding events. The process at each moment of time is in one of $n$ states, and if it is in state number $i$, then it will move to state $j$ with probability $p_{ij}$ (Bertsekas & Tsitsiklis, 2000).
A Markov chain can be represented as a graph in which the vertices are process states and the edges are transitions between states, and on an edge from $i$ to $j$ is written the probability of transition from $i$ to $j$ i.e. $p_{ij}$ ( ).

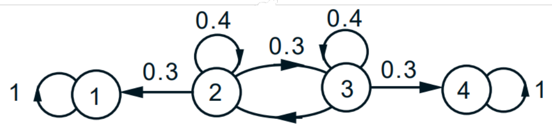


Figure 2.2: Example of Discrete Markov Chain with 4 states

Source: (Bertsekas & Tsitsiklis, 2000).

### 2.6.2 MCMC Sampling

Let's consider we have probability distribution $p(T)$. Monte Carlo methods involve generating a sample from this distribution:

$$T_1, \ldots, T_N \sim p(T).$$

This sample can be used to estimate probability integrals of the form

$$E[f(T)] = \int f(T)p(T)\,dT \approx \frac{1}{N}\sum_{n=1}^{N} f(T_n). \tag{2.10}$$

This allows us to obtain approximate values of the expected function values.
Furthermore, this sample can be used to estimate the mode of the distribution $p(T)$:

$$\max_T p(T) \approx \max_n p(T_n), \tag{2.11}$$

since the occurrence of sample points is most likely in regions of high density values.This property can play an important role in classification tasks, because the mode can indicate the most likely class to which an observation belongs.In Markov Chain Monte Carlo (MCMC) methods, a Markov chain is introduced with a prior distribution $p_0(T)$ and transition probabilities at time $n$, $q_n(T_{n+1}|T_n)$. The generation of samples occurs as follows:

$$\begin{aligned} T_1 &\sim p_0(T), \\ T_2 &\sim q_1(T_2|T_1), \\ &\vdots \\ T_N &\sim q_{N-1}(T_N|T_{N-1}). \end{aligned}$$

By this approach, the generated sample is not a set of independent random variables. However, it is suitable for estimation of probability integrals of the form (4.1) or estimation of the mode of the distribution(Vetrov & Kropotov, 2011).

### 2.6.3 Hamiltonian Monte Carlo

Hamiltonian Monte Carlo (HMC) is a Markov chain Monte Carlo (MCMC) algorithm designed to sample efficiently from complex distributions, especially in high-dimensional spaces. Unlike many MCMC methods that can struggle with random walk behavior or issues with correlated parameters, HMC leverages gradient information to make more directed moves in parameter space. This approach enables it to converge faster to the target distribution compared to simpler techniques like random walk Metropolis or Gibbs sampling.

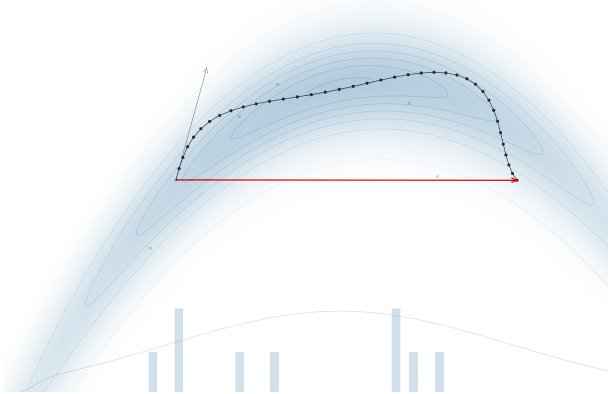

Figure 2.3: Hamilton Markov Chain Sampling

Source:
Feng, C. 2024. *MCMC Interactive Demo.*

However, HMC's efficiency depends on tuning two specific parameters: step size ($\epsilon$) and the number of steps (L). If L is set too low, the algorithm behaves more like a random walk, but if L is too high, it wastes computational resources.

The No-U-Turn Sampler (NUTS) is an extension of HMC that overcomes the challenge of setting L by automatically determining the number of steps needed. NUTS employs a recursive process to build a series of sample points that cover the distribution effectively, halting when it detects the trajectory is turning back on itself. This adjustment allows NUTS to perform as efficiently as a finely tuned HMC and often more efficiently, without manual tuning. NUTS also includes an adaptive step size adjustment, making it ideal for applications that require automated, efficient sampling without extensive user setup, such as automatic inference engines.(Hoffman & Gelman, 2014)

# 3 Logistic Regression

## 3.1 Odds

A fundamental concept in statistical modeling of binary outcomes is the notion of odds. In probability theory and statistics, odds serve as a crucial measure for quantifying the likelihood of events, particularly in contexts where we need to compare the probability of occurrence versus non-occurrence. Formally, odds are defined as the ratio of the probability that an event occurs to the probability that it does not occur (Johnson et.al., 2022):

$$\text{Odds}(C) = \frac{\pi}{1-\pi}, \tag{3.1}$$

where $\pi = P(C)$ represents the probability of event C.

This relationship can be inversely expressed by solving for the probability $\pi$, yielding:

$$\pi = \frac{\text{Odds}(C)}{1 + \text{Odds}(C)}. \tag{3.2}$$

This transformation between probabilities and odds plays a central role in our subsequent development of the logistic regression model.

**Interpreting odds** To develop an intuitive understanding of odds, consider an event with probability $\pi \in [0, 1]$ and its corresponding odds $\frac{\pi}{1-\pi} \in [0, \infty)$. The relationship between odds and probability provides a natural framework for interpreting uncertainty:

- When the odds are less than 1, this indicates the event is more likely to not occur, corresponding to $\pi < 0.5$
- When the odds equal 1, this represents perfect uncertainty with equal probabilities of occurrence and non-occurrence ($\pi = 0.5$)
- When the odds exceed 1, this suggests the event is more likely to occur than not, with $\pi > 0.5$

## 3.2 Logistic Regression Model

Logistic Regression is one of the most important analytical tool in the social and natural sciences and is used for binary classification which involves predicting the probability of one of two possible outcomes. The main difference from classical regression is that logistic regression designed for categorical outcomes (1/0 , Yes/No). Generally we can describe this model as (Jurafsky & Martin, 2024):

$$z = \beta \times x + \epsilon, \tag{3.3}$$

where:
$z$— expresses the weighted sum of the evidence for the class. However, since weights are real-valued, the value of z could be between ($-\infty$ to $+\infty$).Therefore, to obtain probability we should pass $z$ thorough **sigmoid function** (Jurafsky & Martin, 2024) :

$$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+\exp(-z)}. \tag{3.4}$$

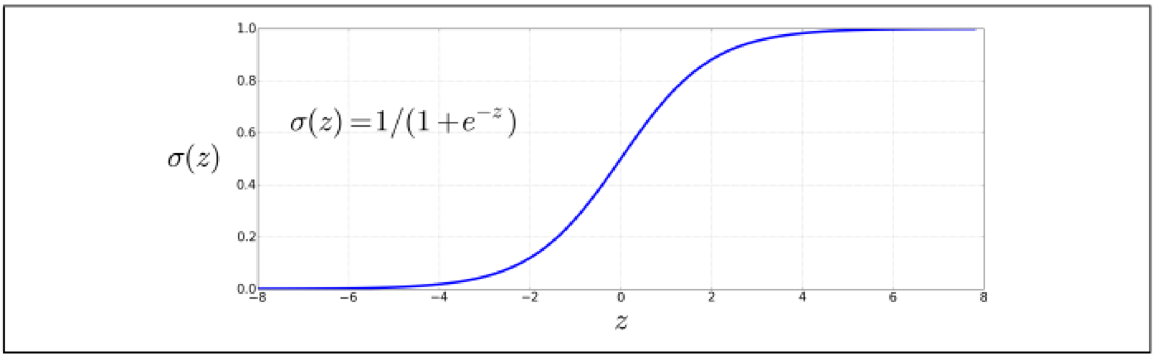


Figure 3.1: Sigmoid function

Source: Jurafsky & Martin, 2024.

$\epsilon$ — the weight vector (coefficients) that shows the significance of each feature for the prediction. Each weight $w_i$ contains information about how strongly this feature influences the probability of the target event.

$x$ — the feature vector, that is, the observed data for each object. Each element of the vector $x_i$ represents one dimension of a feature (for example, age, income, distance, etc.).

$b$ — the bias or intercept, which is added to the linear combination. This shifting value allows the model to predict a non-zero probability of an event even when all feature values are zero.

## 3.3 Maximum Likelihood Estimation

From 5.1 followes that Logistic Regression model can be represented as ( Johnson, Ott,& Dogucu,2022):

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k, \tag{3.5}$$

then:

$$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-\sum \beta_i x_i}}. \tag{3.6}$$

Finally probability of Y=1 for given X :

$$P_i = P(Y = 1 \mid \mathbf{X}) = \frac{1}{1+e^{-\sum_{j=0}^{p} \beta_j x_{ij}}} = \frac{e^{\sum_{j=0}^{p} \beta_j x_{ij}}}{1+e^{\sum_{j=0}^{p} \beta_j x_{ij}}}, \tag{3.7}$$

Where the vector $\mathbf{X} = (x_{i0}, x_{i1}, \ldots, x_{ip})$, with $Y = 1$ and the coefficients $\beta$ are unknown. The logarithm of the odds ratio $\frac{\pi_i}{1-\pi_i}$ is defined as a linear function of $x_{ij}$.

$$\operatorname{logit}(\pi_i) = \ln\left(\frac{\pi_i}{1-\pi_i}\right) = \sum_{j=0}^{p} \beta_j x_{ij}. \tag{3.8}$$

Since weights $\beta$ of our model are unknown we estimate them using method of Maximum Likelihood Estimation (MLE) based on the likelihood function of the selection. For discrete random variables from the probability function $p(x_i, \theta)$, where the probabilities are independent, the likelihood function $L(\theta)$ will be formulated as the probability of the simultaneous occurrence of all elements in the sample:

$$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta). \tag{3.9}$$

Here the log-likelihood function, is commonly used to simplify calculations:

$$\ell(\theta) = \log L(\theta).$$

Next we need to find maximum of function (1.7) by taking the derivative of the function and solving the equation:

$$\frac{\partial \ln L(\theta)}{\partial \theta_j} = \sum_{i=1}^{n} \frac{\partial \ln p(x_i, \theta)}{\partial \theta_j} = 0. \tag{3.10}$$

Having established the necessary framework, we can now proceed to determine the coefficients of logistic regression. Let $y_1, y_2, \ldots, y_n$ denote $n$ observations, each of which can assume binary values of either 0 or 1. The maximum likelihood function is derived from the binomial distribution, incorporating a success indicator, where $y_i = 1$ signifies a success and $y_i = 0$ signifies a failure. The likelihood function for the parameters $\beta$ is articulated as follows in equation (1.10):

$$L(\beta_0, \beta_1, \dots, \beta_p) = \prod_{i=1}^{n} P_i^{y_i}(1-P_i)^{1-y_i} = \prod_{i=1}^{n} \exp\left(y_i \sum_{j=0}^{p} b_j x_{ij}\right) \prod_{i=1}^{n} \left[1 + \exp\left(\sum_{j=0}^{p} b_j x_{ij}\right)\right]^{-1}. \tag{3.11}$$

Next, we take the logarithm of the likelihood function, as shown in equation (1.11):

$$l(\beta_0, \beta_1, \dots, \beta_p) = \ln L = \sum_{i=1}^{n} \sum_{j=0}^{p} b_j x_{ij} y_i - \sum_{i=1}^{n} \ln \left[1 + \exp\left(\sum_{j=0}^{p} b_j x_{ij}\right)\right]. \tag{3.12}$$

The maximum likelihood estimator $\beta$ is obtained by maximizing the logarithm of the likelihood function presented above. The coefficients can be estimated by solving the following $p$ equations derived from equation (1.12), where $p$ represents the number of coefficients $\beta$ :

$$\sum_{i=1}^{n} x_{ij} y_i - \sum_{i=1}^{n} \frac{x_{ij} \exp\left(\sum_{j=0}^{p} b_j x_{ij}\right)}{1 + \exp\left(\sum_{j=0}^{p} b_j x_{ij}\right)} = 0. \tag{3.13}$$

The maximum likelihood estimator $\beta$ is obtained by maximizing the logarithm of the likelihood function presented above. The coefficients can be estimated by solving the following $p$ equations derived from equation (1.12), where $p$ represents the number of coefficients $\beta$ (Lee, 2003):

$$\sum_{i=1}^{n} x_{ij} y_i - \sum_{i=1}^{n} \frac{x_{ij} \exp\left(\sum_{j=0}^{p} b_j x_{ij}\right)}{1 + \exp\left(\sum_{j=0}^{p} b_j x_{ij}\right)} = 0. \tag{3.14}$$

## 3.4 Interpretation of Coefficients

The interpretation of coefficients in logistic regression can be less intuitive than in linear regression because the results indicate how changes in features affect the **log odds** (the logarithm of the ratio of the probability of a positive outcome to the probability of a negative outcome) of the target event.

$$\pi(Y = 1|X) = e^{\sum_{j=0}^{p} \beta_j x_{ij}}. \tag{3.15}$$

From the formula above, we can interpret only the sum of the coefficients. To isolate the partial weight of each feature, the following operation can be done:

$$e^{\beta_j x_{ij}} = \frac{e^{\sum_{j=0}^{p} \beta_j x_{ij}}}{e^{\left(\sum_{j=0}^{p} \beta_j x_{ij} - \beta_j x_{ij}\right)}}. \tag{3.16}$$

After applying the natural logarithm, we get:

$$\ln(e^{\beta_j x_{ij}}) = \beta_j x_{ij}. \tag{3.17}$$

This allows us to interpret $\beta_j$ as the effect of the feature $x_i$ on the log odds of the target event.

- If $e^{\beta_i} > 1$, an increase in the feature raises the odds of the target event.
- If $e^{\beta_i} < 1$, an increase in the feature lowers the odds of the target event.
- If $e^{\beta_i} = 1$, the feature does not influence the odds of the target event.

## 3.5 Interpretation Example

Suppose we have a model predicting the probability of heart disease, and the coefficient for the feature

$$e^{1.2} \approx 3.32,$$

which means that men have approximately 3.32 times higher odds of having heart disease compared to women.

### Interpretation for Continuous and Categorical Features

- **Continuous features**: The coefficient indicates the change in log odds with a one-unit increase in the feature.
- **Categorical features**: The coefficient is interpreted relative to a reference category (e.g., for a binary feature *Gender*, the coefficient shows the change in odds for men compared to women, if the reference category is women).

## 3.6 Interval Estimations

An important part of this bachelor's thesis is confidence intervals, or, as they are called in Bayesian statistics, credible intervals. They are used to estimate the range of parameter values that, with a given probability (the most popular one are 90% , 95% or 99% as mentioned (Malá, 2013)), are likely to contain the true parameter. Credible intervals provide a more intuitive representation of the uncertainty of model parameters compared to classical confidence intervals. Unlike the frequentist approach, where the interval is calculated based on hypothetical repeated samples, the Bayesian approach utilizes the posterior distribution of parameters, derived from real data and specified prior assumptions.

**Confidence Intervals — Classical Approach** In the classical approach, a confidence interval replaces a point estimate with a specifically constructed range of values that is highly likely to contain the unknown parameter. This likelihood is referred to as the confidence level and is denoted as $1 - \alpha$, where $\alpha \in (0; 1)$. The interval, bounded by a lower limit ($\theta_D$) and an upper limit ($\theta_H$), is called a confidence interval. It is typically constructed using the point estimate and its transformation.

For a two-sided confidence interval, the probability is distributed symmetrically:

$$\mathbb{P}(\theta_D < \theta < \theta_H) = 1 - \alpha, \tag{3.18}$$

which implies

$$\mathbb{P}(\theta < \theta_D) = \frac{\alpha}{2}, \quad \mathbb{P}(\theta > \theta_H) = \frac{\alpha}{2}. \tag{3.19}$$

For one-sided intervals, the probabilities are expressed as

$$\mathbb{P}(\theta_D < \theta) = 1 - \alpha \quad \text{(left-sided)}, \tag{3.20}$$

$$\mathbb{P}(\theta > \theta_H) = 1 - \alpha \quad \text{(right-sided)}. \tag{3.21}$$

An example of a classical confidence interval is the estimation of the mean of a normal distribution when the variance is known. In this case, the statistic $\overline{X}$ follows the normal distribution $\mathcal{N}(\mu, \sigma^2/n)$. Using the standardized normal distribution and quantiles, the interval can be written as:

$$\left(\overline{X} - u_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \overline{X} + u_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right). \tag{3.22}$$

Once a sample is taken, the interval becomes fixed, and the probability interpretation reflects the frequency with which this method captures the true parameter value across repeated samples. (Malá, 2013)

**Credible Intervals — Bayesian Approach** The Bayesian approach to intervals relies on the posterior distribution, which encapsulates all information about the unknown parameter. Since the parameter is treated as a random variable, the Bayesian interval, referred to as the credible interval, is defined as a range of values where the parameter lies with a specified probability. For a parameter $\theta$, the interval is determined by the condition:

$$\mathbb{P}(\theta \in C) = 1 - \alpha, \tag{3.23}$$

where $C$ is the region containing the parameter with the specified probability.

To find the most accurate interval, the highest posterior density (HPD) region is often used. This region $C^*$ satisfies the following conditions:

1. $\mathbb{P}(\theta \in C^*) = 1 - \alpha$,

2. For all $\theta_1 \in C^*$ and $\theta_2 \notin C^*$, the inequality

$$g(\theta_1|y) \geq g(\theta_2|y) \tag{3.24}$$

   holds, where $g(\theta|y)$ is the posterior density.

For unimodal distributions, the HPD region is typically a single interval with bounds found by solving:

$$\int_{-\infty}^{\theta_D} g(\theta|y)d\theta = \frac{\alpha}{2}, \quad \int_{\theta_H}^{\infty} g(\theta|y)d\theta = \frac{\alpha}{2}. \tag{3.25}$$

For more complex distributions, the bounds of the HPD region can be found numerically. The algorithm usually starts by identifying the mode of the posterior density and iteratively refining the bounds to meet the chosen probability level. The HPD region accounts for the entire structure of the distribution, providing the most precise and informative interval estimate.

Thus, logistic regression coefficients provide information about the strength and direction of a feature's effect on the likelihood of the target event. Koop(2007)

## 3.7 Measures of the performance

After fitting Logistic Regression, it's essential to evaluate how well the model performs on data. For this purpose, several performance metrics help assess accuracy, class distinction capability, and prediction quality. Let's review the main metrics that can be used:

### 3.7.1 Confusion Matrix

In the case of binary classification, the confusion matrix is a table consisting of two rows and two columns, where the rows correspond to the actual classes and the columns represent the predicted ones. During the training process, the classifier makes predictions

| | Predicted Positive | Predicted Negative |
|---|---|---|
| Actual Positive | True Positive (TP) | False Negative (FN) |
| Actual Negative | False Positive (FP) | True Negative (TN) |

Table 1: Confusion Matrix

on training examples for which the class label is known. It also makes errors of Type I and Type II. If the predicted class matches the actual class, the classification outcome is considered true; otherwise, it is considered false. Examples of the positive and negative classes where the prediction outcome is true are called true positives (TP) and true negatives (TN), respectively. These are correctly classified examples.

Examples of the positive and negative classes where the prediction outcome is false are called false positives (FP) and false negatives (FN), respectively. These are cases where the classifier has made an error.

To assess the quality of a model's performance, a range of indicators is used, some of which are based on the analysis of elements from the confusion matrix. These indicators can include Accuracy, Precision, Recall, F1 score, or ROC AUC. Each of these indicators provides a different perspective on the effectiveness of the classification model.

### 3.7.2 Accuracy

Definitions of all metrics were taken from (Machine Learning Crash Course,2024).

Accuracy is a metric used to evaluate the performance of a classification model. It measures the proportion of correctly predicted instances out of the total instances in the dataset. It can be calculated using the formula:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{3.26}$$

Essentially, this metric expresses the ratio of all correctly estimated cases to the total number of observations, regardless of which group they belong to. Accuracy is a criterion that characterizes the quality of the model aggregated across all classes. It is useful when the classes hold equal significance for us. However, if this is not the case, accuracy can be misleading.

### 3.7.3 Precision

Precision is a metric used to evaluate the performance of a classification model, particularly in binary classification. It measures the proportion of true positive predictions among all positive predictions made by the model. Precision can be calculated using the formula:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{3.27}$$

High precision indicates that positive predictions are likely to be correct, which is crucial in scenarios where false positives have significant consequences.

### 3.7.4 Recall

The true positive rate (TPR), or the proportion of all actual positives that were classified correctly as positives, is also known as recall.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{3.28}$$

In other words, sensitivity indicates how many of the actual successful cases were correctly identified. We can say that sensitivity is the percentage of correctly predicted successes in our model. An ideal model would result in no false negatives, achieving a recall (TPR) of 1.0, which corresponds to a detection rate of 100%.

However, in situations where the dataset is imbalanced and contains only a very small number of actual positive instances—perhaps just 1 or 2—recall becomes a less significant and less effective metric.

### 3.7.5 F1-Score

The F1 score is a metric that evaluates the performance of a classification model, especially useful for imbalanced datasets. It combines precision and recall, providing a balanced view of the model's effectiveness.In this context, the F1 score serves as a universal tool for assessing the quality of classification models.

$$\text{F-1 score } = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.29}$$

The F1 score ranges from 0 to 1, where 1 represents perfect precision and recall, while 0 indicates poor performance. In the context of this work, the F1 score will be more significant than other metrics.

### 3.7.6 ROC-AUC curve

The ROC curve was first developed by electrical engineers and radar engineers during World War II for detecting enemy objects in battlefields, starting in 1941, which led to its name "receiver operating characteristic". Now it is used as a visual representation of model performance across all thresholds.(Junge & Dettori, 2024)

An ROC curve works by plotting the true positive rate (TPR) on the y-axis and the false positive rate (FPR) on the x-axis of a graph.

$$\text{TPR} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}. \tag{3.30}$$

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}. \tag{3.31}$$

An ideal classifier would maximize the TPR while minimizing the FPR, resulting in a curve that quickly rises toward the top left corner of the plot. The **Area Under the ROC Curve (AUC)** provides a summary measure of the model's discriminative ability, with an AUC close to 1 indicating excellent classification performance and an AUC close to 0.5 suggesting performance close to random chance.

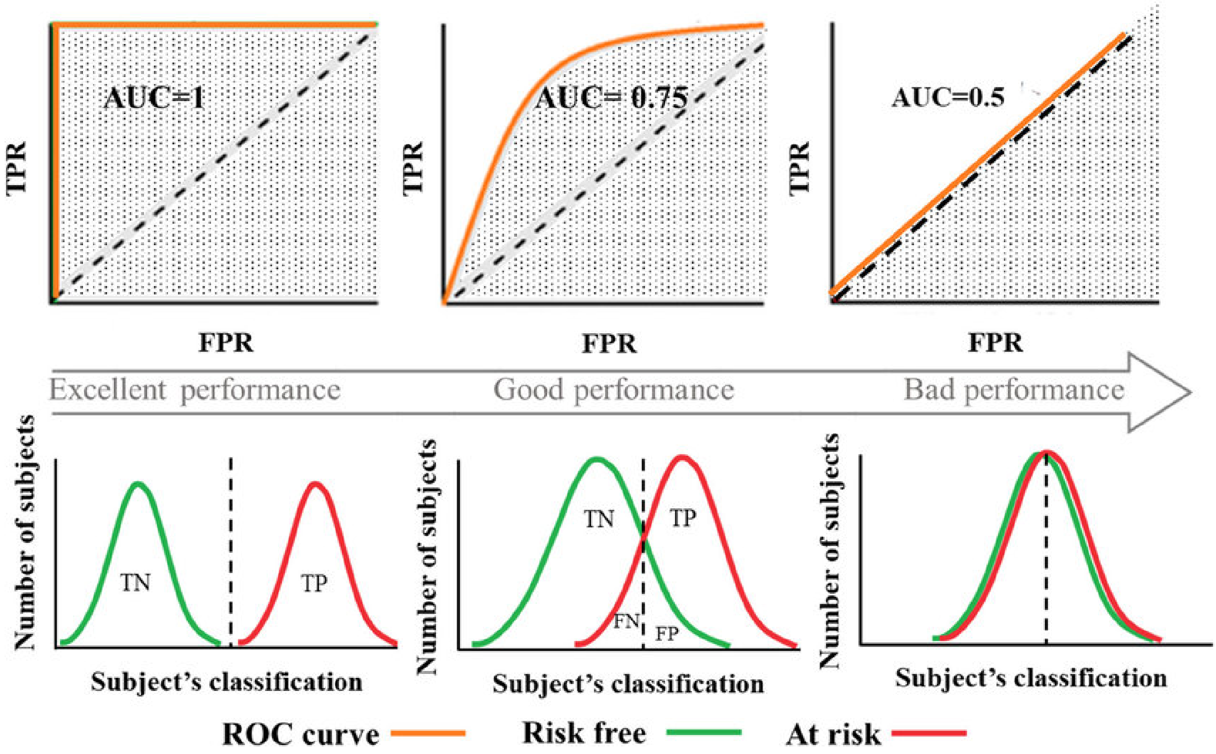


Figure 3.2: llustration-of-AUC-classification-metric

Source: Aldraimli,2020.

In practice, the ROC curve is valuable for comparing classifiers across various threshold values, enabling the selection of an optimal threshold based on the desired balance between true positives and false positives.

Another question we would like to answer is what AUC is considered good. In practice, the threshold is not entirely clear and can be ambiguous. For example, in fields such as medicine, an extremely high AUC around 0.95 is required, while in marketing, by contrast, we can afford to sacrifice some accuracy, and an AUC of 0.8 would be considered a good result.

However, we can make a generalization:
For example, in the book (Hosmer & Lemeshow, 2000), authors propose the following classification, which has become somewhat of a standard:

- $AUC = 0.5$: This indicates no discrimination, and such a result is comparable to random coin tossing.
- $0.5 < AUC < 0.7$: This value is considered poor discrimination, which is not significantly better than random coin tossing.
- $0.7 \leq AUC < 0.8$: This value is regarded as acceptable discrimination.
- $0.8 \leq AUC < 0.9$: This value is considered excellent discrimination.
- $AUC \geq 0.9$: This value is regarded as outstanding discrimination.

# 4 Empirical Analysis

## 4.1 Exploratory Data Analysis and Preprocessing

In this thesis, I used publicly available data from the UC Irvine Machine Learning Repository. The data consists of a CSV file with 13 variables representing features and 918 observations. The objective is to build a model based on these features to predict whether a patient has heart disease. This dataset is well-suited for the primary goal of this work — to explore the application of logistic and Bayesian logistic regression methods on real-world data — due to its range of intuitively understandable features, such as Age, Cholesterol levle, Resting Blood Pressure, Max Heart Rate Achieved.

### 4.1.1 Description of Dataset

For better understanding, I will divide the dataset based on variable types. There are four types: ordinal, binary, categorical, and quantitative.

### 4.1.2 Quantitative Variables

- **Age**: the patient's age, which is a key risk factor for cardiovascular diseases, as the likelihood of developing heart disease increases with age.
- **Resting BP**: the patient's resting blood pressure (measured in mm Hg). High blood pressure is a known risk factor for cardiovascular diseases, as it can lead to blood vessel and heart muscle damage.
- **Cholesterol**: blood cholesterol level (measured in mg/dl). High cholesterol levels can lead to atherosclerotic plaque buildup, increasing the risk of heart disease.
- **MaxHR**: the maximum heart rate achieved by the patient during physical exertion. High values may indicate good physical fitness, while lower values can suggest potential heart issues.
- **Oldpeak**: ST depression induced by exercise relative to rest (measured in mm). This parameter measures the ST segment shift on an electrocardiogram (ECG) and may signal myocardial ischemia, which is important for diagnosing cardiovascular diseases.

### 4.1.3 Nominal Variables

- **ChestPainType**: type of chest pain experienced by the patient, with categories such as typical angina, atypical angina, non-anginal pain, or asymptomatic. These categories help clinicians assess the risk of coronary artery disease.
- **RestingECG**: resting electrocardiogram results, which may include normal, ST-T wave abnormality, or left ventricular hypertrophy. These categories help evaluate heart health and potential abnormalities.
- **STSlope**: slope of the ST segment on the ECG during exercise, which can be upsloping, flat, or downsloping. This parameter is crucial for assessing ischemia risk, as a downsloping ST segment can be indicative of significant heart issues.

### 4.1.4 Binary Variables

- **Sex**: the patient's gender, where "1" typically denotes male and "0" denotes female. Gender can influence cardiovascular risk factors and may affect the interpretation of other parameters.
- **FastingBS**: fasting blood sugar level, where "1" indicates an elevated level ($\geq 120$ mg/dl) and "0" indicates a normal level. Elevated blood sugar can be a sign of diabetes, which increases the risk of heart disease.

- **ExerciseAngina**: presence of exercise-induced angina, where "1" indicates chest pain during physical activity and "0" indicates no pain. This parameter is important for diagnosing ischemic heart disease, as chest pain with exercise may be related to inadequate blood flow to the heart.

| Age | Sex | ChestPainType | RestingBP | Cholesterol | FastingBS | RestingECG | MaxHR | ExerciseAngina | Oldpeak | ST_Slope | HeartDisease |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 40 | M | ATA | 140 | 289 | N | Normal | 172 | N | 0.0 | Up | 0 |
| 49 | F | NAP | 160 | 180 | N | Normal | 156 | N | 1.0 | Flat | 1 |
| 37 | M | ATA | 130 | 283 | N | ST | 98 | N | 0.0 | Up | 0 |
| 48 | F | ASY | 138 | 214 | N | Normal | 108 | Y | 1.5 | Flat | 1 |
| 54 | M | NAP | 150 | 195 | N | Normal | 122 | N | 0.0 | Up | 0 |

Table 2: Sample data

### 4.1.5 Correlation Analysis

For further work with the dataset, it is important to ensure that there is no multicollinearity in the model and that all variables are generally linearly independent.

Initially, we will check for multicollinearity by calculating the correlation coefficients between the variables. These coefficients range from -1 to 1. For this analysis, we will use Spearman's rank correlation coefficient ($\rho$), which is defined as follows:

$$\rho = 1 - \frac{6\sum_{i=1}^{n}(p_i - q_i)^2}{n(n^2-1)}, \tag{4.1}$$

where $n$ is the number of observations, $p_i$ is the rank of variable $X$, and $q_i$ is the rank of variable $Y$. $X$ and $Y$ are the two variables that have been ranked according to their sizes. In cases where the two variables have values close to an absolute value of 1, it means that the variables are strongly correlated, and one of the quantities is a monotonic function of the other(Bílková, 2009).

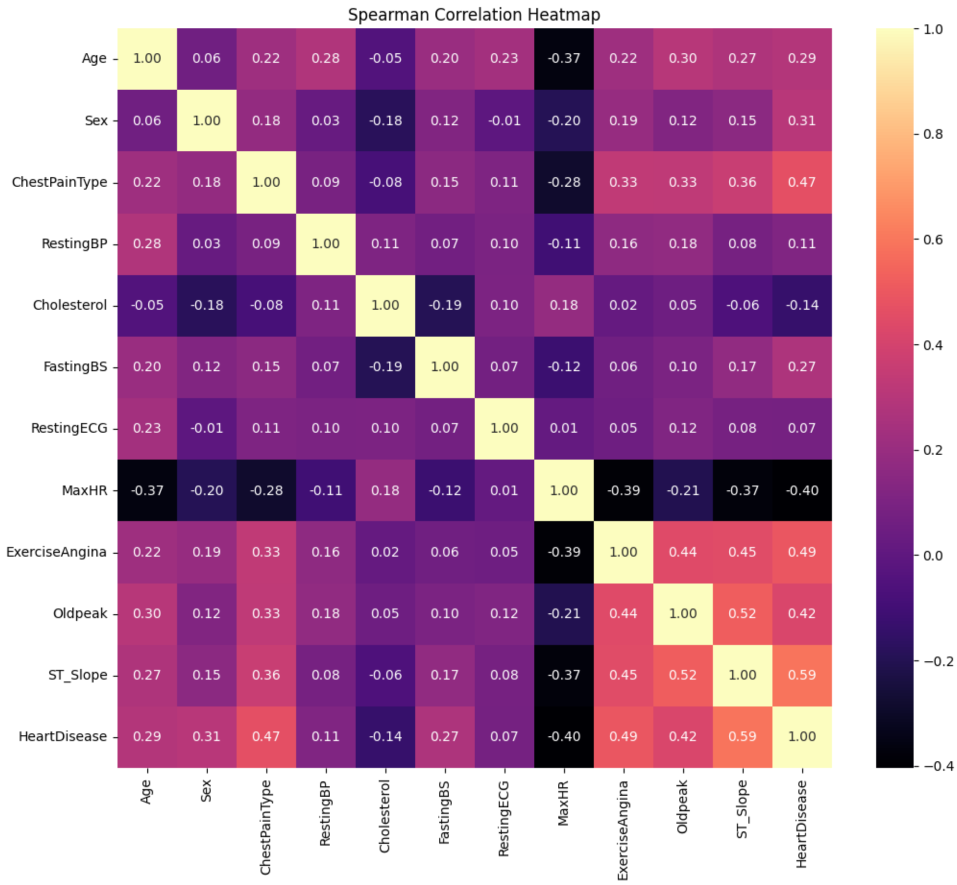


Figure 4.1: Spearman Correlation Matrix

Source: Own processing in Python

Next, we will examine the Pearson correlation coefficient ($r$), which measures the strength and direction of the linear relationship between two continuous variables. The Pearson correlation coefficient is calculated as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}, \tag{4.2}$$

where $\bar{x}$ and $\bar{y}$ are the sample means of $X$ and $Y$, respectively. Like Spearman's coefficient, Pearson's $r$ ranges from -1 to +1, where:

- $r = +1$ indicates a perfect positive linear correlation,
- $r = -1$ indicates a perfect negative linear correlation,
- $r = 0$ indicates no linear correlation.

The Pearson correlation coefficient is based on several key assumptions:

1. **Level of Measurement**: Variables should be measured at the interval or ratio level
2. **Linear Relationship**: There should be a linear relationship between the variables
3. **Bivariate Normality**: Variables should be approximately normally distributed
4. **Homoscedasticity**: The variability in scores should be roughly the same at all values of the other variable

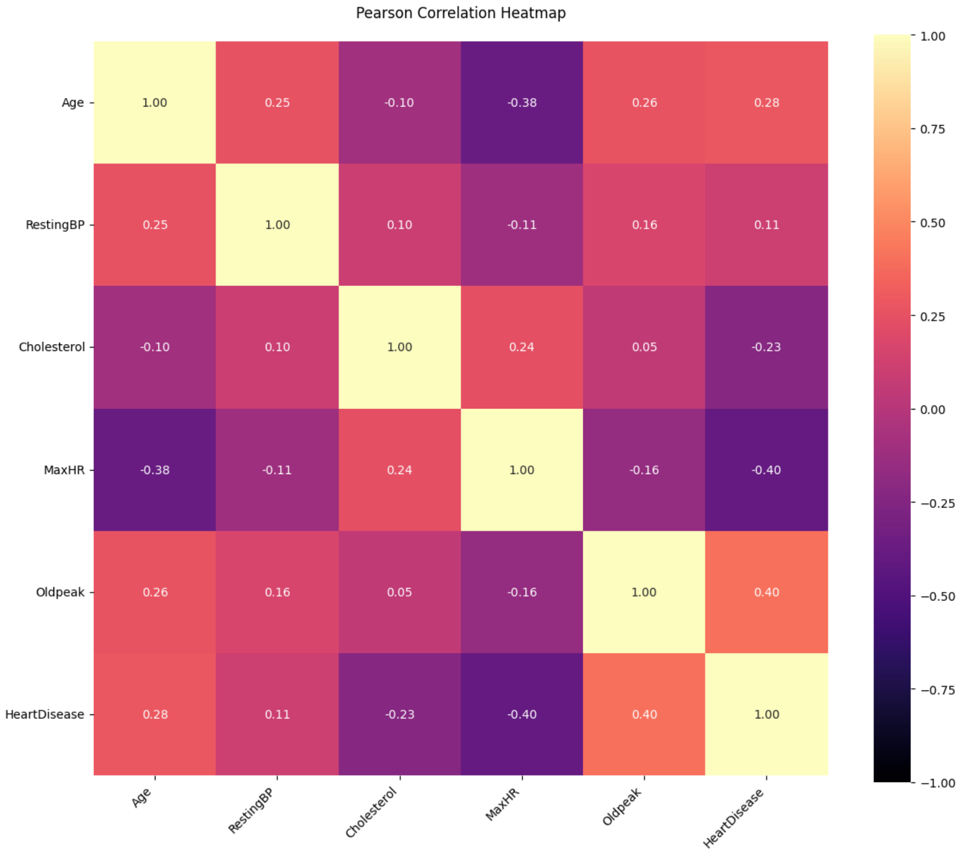


Figure 4.2: Pearson Correlation Matrix

Source: Own processing in Python

Comparing the results of both Spearman's ($\rho$) and Pearson's ($r$) correlation matrices, we can observe that the correlation patterns are quite similar, which suggests consistency in the relationships between variables regardless of whether we consider rank-based (Spearman) or linear (Pearson) correlations. The highest correlation coefficients in both matrices do not exceed 0.5, indicating the absence of strong multicollinearity among the predictor variables.(Bílková,2007)

This consistency between Spearman's and Pearson's correlations also suggests that the relationships between variables are relatively stable and not significantly affected by outliers or non-linear patterns. The lack of strong correlations ($|r| > 0.7$) between independent variables indicates that each variable likely contributes unique information to the model, which is desirable for building a robust predictive model. It should be noted that the choice of boundary is subjective and varies from source to source.

The practical implications of these correlation analyses are:

- **Model Stability**: Low correlations between predictors suggest that the model estimates will be stable.
- **Variable Importance**: Each variable provides unique information, making it easier to interpret individual effects.
- **Prediction Accuracy**: The absence of multicollinearity reduces the risk of overfitting and improves the model's predictive performance.

Therefore, based on both correlation analyses, we can conclude that multicollinearity is not a significant concern in our dataset, and all variables can be retained for further analysis without the need for feature elimination based on correlation criteria. This provides a solid foundation for subsequent modeling steps, ensuring that our predictive models will be both stable and interpretable.

Now that we are sure we do not have correlated features, we can begin encoding the data.

### 4.1.6 Data Preprocessing for Modeling

In this study, the original dataset was divided into two parts: the main part (20% of the data, 184 observations) and the auxiliary part (80% of the data, 734 observations). This choice of split ratio has specific reasons. The main part represents a smaller but representative sample, which is used as a source of external information for forming informative priors in logistic regression. This approach simulates a real-world scenario where prior knowledge about the analyzed dataset is available, for instance, from previous studies or historical data.

The 20% to 80% split was chosen to ensure that the main part contains a sufficiently large number of observations to create stable and reliable informative priors without significantly reducing the size of the auxiliary part. The auxiliary part, which constitutes the majority of the data, is then used for estimating the logistic regression model itself, where the informative priors derived from the main part are utilized as prior knowledge.

Specifically, the main part serves as input for estimating the distributions of the model parameters. These estimates provide information about the likely values of coefficients in logistic regression, helping to improve the model's convergence and stability when analyzing the auxiliary part. This approach ensures the effective integration of external knowledge into Bayesian modeling while leaving a sufficiently large dataset for valid estimates and model testing on the auxiliary part.

It is important to note that data preprocessing and train-test split were conducted independently for each subset to avoid data leakage.

### 4.1.7 Data Encoding

To use logistic regression, we must first encode the data, as this classifier cannot directly process text and can only work with numerical variables.

There are many ways to achieve this. For example, we can represent a category variable as an integer that denotes only that specific category. In this case, we must have a clear understanding of what the encoded variable represents, and the correct order. Otherwise, there is a risk of incorrect classification of the category of this variable.

Another method is called binary encoding. It is suitable for nominal and categorical data, but unlike simple text-to-number replacement, it represents each category of a categorical variable as a separate binary variable, where a new column is created for each category, taking the value of 1 if the object belongs to that category, and 0 otherwise. It is important to note that the number of encoded columns is always one less than the number of categories, as it is assumed that when all other categories are zero, the K-th category is equal to 1. Conversely, if one of the categories is non-zero, all other categories, including the K-th one, are zero. The last category is often referred to as the **reference category**. If we violate this encoding rule, the matrix X will become singular, which means there will be perfect multicollinearity among the regressors in our model, leading to incorrect results.

The second encoding method was chosen for the dataset. The main benefit of this method is that it increases the number of coefficients $\beta$ the model needs to estimate to the number of categories K minus 1.This approach helps to avoid the issue of multicollinearity, which could arise if a single numeric variable were used for categorical features.The issue is that if a single variable were used for categorical features, the model might mistakenly interpret these values as quantitative (e.g., 1, 2, 3, and so on), which could create false assumptions about the order or intervals between categories. By applying binary encoding, we ensure that the model treats each category as a separate, independent variable, avoiding such problems and improving model accuracy.

#### 4.1.8 Train-test Split

The final step before applying logistic regression is splitting the data into two sets. One will be used for training the models (80 percent of the data), while the other will be used for testing the classification capabilities of the models (20 percent of the data). In our case, we are dealing with a relatively balanced dataset, where 55 percent of the cases have heart disease and the remaining 45 percent do not, so techniques like oversampling and undersampling will not be used.

#### 4.1.9 Data Normalization

Normalization is a method of transforming data to bring it into a common scale or range of values. This is particularly important when the features in the dataset have different units of measurement or vary significantly in magnitude as it is in our dataset. The goal of normalization is to make the data more suitable for analysis and model training, as well as to eliminate the impact of differences in the scales of variables.

Data normalization plays a crucial role in data preprocessing, as it enhance model performance and speed up the training process. It is especially usefull when using samplers like NUTS and Hamiltonian Monte Carlo, as normalization helps make the parameter space more homogeneous, which contributes to faster and more efficient sampling.

For normalization, MinMaxNormalization method is used, which is represented by the following formula:

$$X' = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{4.3}$$

The advantage of this method is that the normalized values will always lie within a specified range, typically between 0 and 1 or -1 and 1(Jaiswal,2024).

#### 4.1.10 Empirical Estimation of Informative Prior

As mentioned earlier, logistic regression was applied to an auxiliary dataset. The logistic regression model was implemented using the scikit-learn library in Python. All features were retained in the model since preliminary analysis showed no significant multicollinearity among the variables.

Our model looks as follows :

$$\begin{aligned} \text{HeartDisease} \sim\ & \beta_0 + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{Sex} + \beta_3 \cdot \text{RestingBP}+ \\ & \beta_4 \cdot \text{Cholesterol} + \beta_5 \cdot \text{FastingBS} + \beta_6 \cdot \text{MaxHR}+ \\ & \beta_7 \cdot \text{ExerciseAngina} + \beta_8 \cdot \text{Oldpeak} + \beta_9 \cdot \text{ChestPainType\_ATA}+ \\ & \beta_{10} \cdot \text{ChestPainType\_NAP} + \beta_{11} \cdot \text{ChestPainType\_TA}+ \\ & \beta_{12} \cdot \text{RestingECG\_Normal} + \beta_{13} \cdot \text{RestingECG\_ST}+ \\ & \beta_{14} \cdot \text{ST\_Slope\_Flat} + \beta_{15} \cdot \text{ST\_Slope\_Up}. \end{aligned} \tag{4.4}$$

The coefficients are presented in the following table:

The coefficients of logistic regression were estimated using the Newton-Raphson method. This method aims to find parameter estimates that maximize the log-likelihood function. The core idea behind the method is to approximate the function around the current

```
Optimization terminated successfully.
         Current function value: 0.320161
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:           HeartDisease   No. Observations:                  734
Model:                          Logit   Df Residuals:                      718
Method:                           MLE   Df Model:                           15
Date:                Mon, 11 Nov 2024   Pseudo R-squ.:                  0.5352
Time:                        16:58:57   Log-Likelihood:                -235.00
converged:                       True   LL-Null:                       -505.62
Covariance Type:            nonrobust   LLR p-value:                 1.050e-105
=====================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
Intercept            -0.7299      1.600     -0.456      0.648      -3.865       2.405
Age                   0.0149      0.015      0.972      0.331      -0.015       0.045
Sex                   1.3177      0.305      4.325      0.000       0.721       1.915
RestingBP             0.0018      0.007      0.263      0.793      -0.012       0.015
Cholesterol          -0.0050      0.001     -3.951      0.000      -0.007      -0.002
FastingBS             1.0417      0.310      3.357      0.001       0.434       1.650
MaxHR                -0.0038      0.006     -0.659      0.510      -0.015       0.007
ExerciseAngina        1.0986      0.278      3.953      0.000       0.554       1.643
Oldpeak               0.4588      0.132      3.475      0.001       0.200       0.718
ChestPainType_ATA    -1.4961      0.370     -4.039      0.000      -2.222      -0.770
ChestPainType_NAP    -1.5468      0.301     -5.131      0.000      -2.138      -0.956
ChestPainType_TA     -1.1667      0.537     -2.173      0.030      -2.219      -0.114
RestingECG_Normal    -0.2202      0.306     -0.719      0.472      -0.820       0.380
RestingECG_ST        -0.4812      0.393     -1.225      0.221      -1.251       0.289
ST_Slope_Flat         1.3525      0.486      2.785      0.005       0.401       2.304
ST_Slope_Up          -1.0981      0.514     -2.137      0.033      -2.105      -0.091
=====================================================================================
```

Figure 4.3: Regression Parameter Estimates of Frequentist Logistic Regression

Source: Own processing in Python

estimate using a quadratic polynomial and then search for the extreme point of this polynomial, providing a more accurate estimate with each iteration.(Hastie et al.,2009).

Upon examination of the results, many coefficients were found to be statistically insignificant and were subsequently removed from the model. This feature selection process resulted in a refined model containing only statistically significant coefficients. This optimized model specification was then applied to the smaller dataset for consistency in our comparative analysis.

$$\begin{aligned}\text{HeartDisease} \sim \beta_0 + \beta_1 \cdot \text{Sex} + \beta_2 \cdot \text{Cholesterol} + \beta_3 \cdot \text{FastingBS}+ \\ +\beta_4 \cdot \text{ChestPainTypeATA} + \beta_5 \cdot \text{ExerciseAngina} + \beta_6 \cdot \text{ChestPainTypeNAP}+ \\ +\beta_7 \cdot \text{STSlopeFlat}\end{aligned} \tag{4.5}$$

After removing all statistically insignificant variables, we reconstructed the logistic regression model using equation (8.2).

Estimated parameters and their distributions from this model will be used further used as informative priors.

## 4.2 Logistic Regression Model

Now we proceed to the main and most substantial part of the bachelor thesis: building and comparing models on our target data.

```
=== New Logistic Regression Model (Significant Features Only) ===
Optimization terminated successfully.
         Current function value: 0.308533
         Iterations 7

=== Final Model Summary (Significant Features Only) ===
                           Logit Regression Results
==============================================================================
Dep. Variable:           HeartDisease   No. Observations:                  550
Model:                          Logit   Df Residuals:                      542
Method:                           MLE   Df Model:                            7
Date:                Mon, 02 Dec 2024   Pseudo R-squ.:                  0.5490
Time:                        13:47:00   Log-Likelihood:                -169.69
converged:                       True   LL-Null:                       -376.24
Covariance Type:            nonrobust   LLR p-value:                 3.717e-85
=====================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                -2.5824      0.444     -5.817      0.000      -3.453      -1.712
Sex                   1.9501      0.391      4.994      0.000       1.185       2.716
Cholesterol          -0.3732      0.155     -2.415      0.016      -0.676      -0.070
FastingBS             1.3003      0.366      3.551      0.000       0.583       2.018
ExerciseAngina        1.7322      0.311      5.577      0.000       1.123       2.341
ChestPainType_ATA    -1.5433      0.413     -3.737      0.000      -2.353      -0.734
ChestPainType_NAP    -2.0054      0.348     -5.763      0.000      -2.687      -1.323
ST_Slope_Flat         2.7985      0.317      8.820      0.000       2.177       3.420
=====================================================================================
```

Figure 4.4: Regression Parameter Estimates of Frequentist Logistic Regression with only statistically significant features

Source: Own processing in Python

### 4.2.1 Regression Parameter Estimates of Frequentist Logistic Regression

```
=== New Logistic Regression Model (Significant Features Only) ===
Optimization terminated successfully.
         Current function value: 0.273174
         Iterations 8

=== Final Model Summary (Significant Features Only) ===
                           Logit Regression Results
==============================================================================
Dep. Variable:           HeartDisease   No. Observations:                  137
Model:                          Logit   Df Residuals:                      129
Method:                           MLE   Df Model:                            7
Date:                Mon, 02 Dec 2024   Pseudo R-squ.:                  0.5991
Time:                        16:51:15   Log-Likelihood:                -37.425
converged:                       True   LL-Null:                       -93.345
Covariance Type:            nonrobust   LLR p-value:                 3.810e-21
=====================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                -2.5960      0.995     -2.610      0.009      -4.545      -0.647
Sex                   2.4241      0.915      2.649      0.008       0.631       4.217
Cholesterol          -0.4420      0.366     -1.207      0.227      -1.160       0.276
FastingBS             1.5667      0.805      1.947      0.051      -0.010       3.144
ExerciseAngina        2.2390      0.717      3.121      0.002       0.833       3.645
ChestPainType_ATA    -3.8981      1.099     -3.546      0.000      -6.053      -1.743
ChestPainType_NAP    -1.8851      0.678     -2.781      0.005      -3.214      -0.556
ST_Slope_Flat         2.3815      0.738      3.229      0.001       0.936       3.827
=====================================================================================
```

Figure 4.5: Regression Parameter Estimates of Frequentist Logistic Regression



Source: Own processing in Python

We obtained a coefficient table for the smaller dataset. Although one variable showed statistical insignificance, we decided to retain it in the model since our analysis of the larger dataset indicates that this variable becomes significant with a larger sample size.

### 4.2.2 Confusion Matrix

Confusion matrix is as a key tool for evaluating model effectiveness. This matrix compares the actual values of the target variable with the model's predicted results. It is particularly important to note that the presented results reflect the model's performance on test data that was not involved in the training process, thus providing an objective assessment of its real predictive capability.

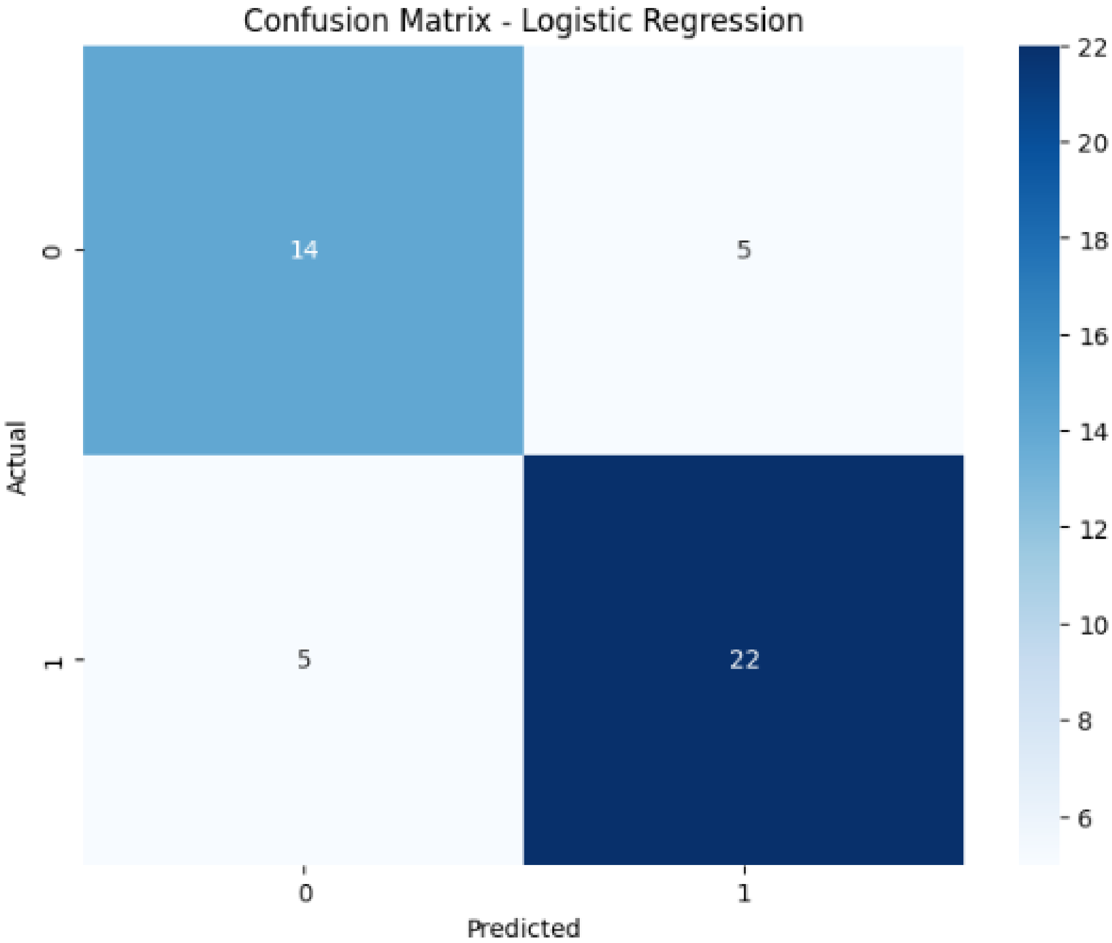


Figure 4.6: Confusion Matrix of Frequentist Logistic Regression Model

Source: Own processing in Python

As we can see, the model performs quite well with class 1 and slightly worse with class 0.To be more precise, it correctly classifies 22 people as having heart disease and 14 as healthy. At the same time, 5 patients were misclassified as having heart disease when they were actually healthy, and 5 were misclassified as healthy, although they actually had heart disease.

### 4.2.3 Metrics of the Model

Next, from this table, we can derive the following metrics:

Table 3: Model Performance Metrics

| Metric | Value |
|---|---|
| Accuracy | 0.78 |
| Precision | 0.81 |
| Recall | 0.81 |
| F1-score | 0.81 |
| ROC AUC | 0.81 |

Source: Own processing in Python

- **Accuracy (0.78)**: This is the overall rate of correct predictions made by the model across all classes. An accuracy of 0.78 means the model correctly predicts 0.78% of all cases.
- **Precision (0.81)**: Precision, or the positive predictive value, measures the proportion of positive predictions that are actually correct. A precision of 0.81 means that, among all instances predicted as positive, 0.81% are true positives.
- **Recall (0.81)**: Also known as sensitivity or true positive rate, recall shows the percentage of actual positives the model correctly identified. A recall of 0.81 means the model captures 0.81% of the actual positive cases.
- **F1-score (0.81)**: The F1-score is the harmonic mean of precision and recall. An F1-score of 0.81suggests the model maintains a good balance between these two metrics.
- **ROC AUC (0.81)**: The ROC AUC (Receiver Operating Characteristic Area Under Curve) measures the model's ability to distinguish between classes, regardless of classification threshold. An AUC of 0.81 indicates the model performs very well in distinguishing between classes.

#### 4.2.4 ROC-curve

Another important model quality metric is the ROC curve. This curve helps visualize the model's performance by measuring its ability to distinguish between classes, and in some cases, it can also signal overfitting. The diagonal represents a random classifier, meaning a model that randomly guesses between the classes. If a model's ROC curve is close to this diagonal, it indicates poor performance, as the model is not able to distinguish between the classes better than random guessing. On the other hand, a ROC curve that moves towards the top-left corner (high true positive rate and low false positive rate) indicates a better-performing model

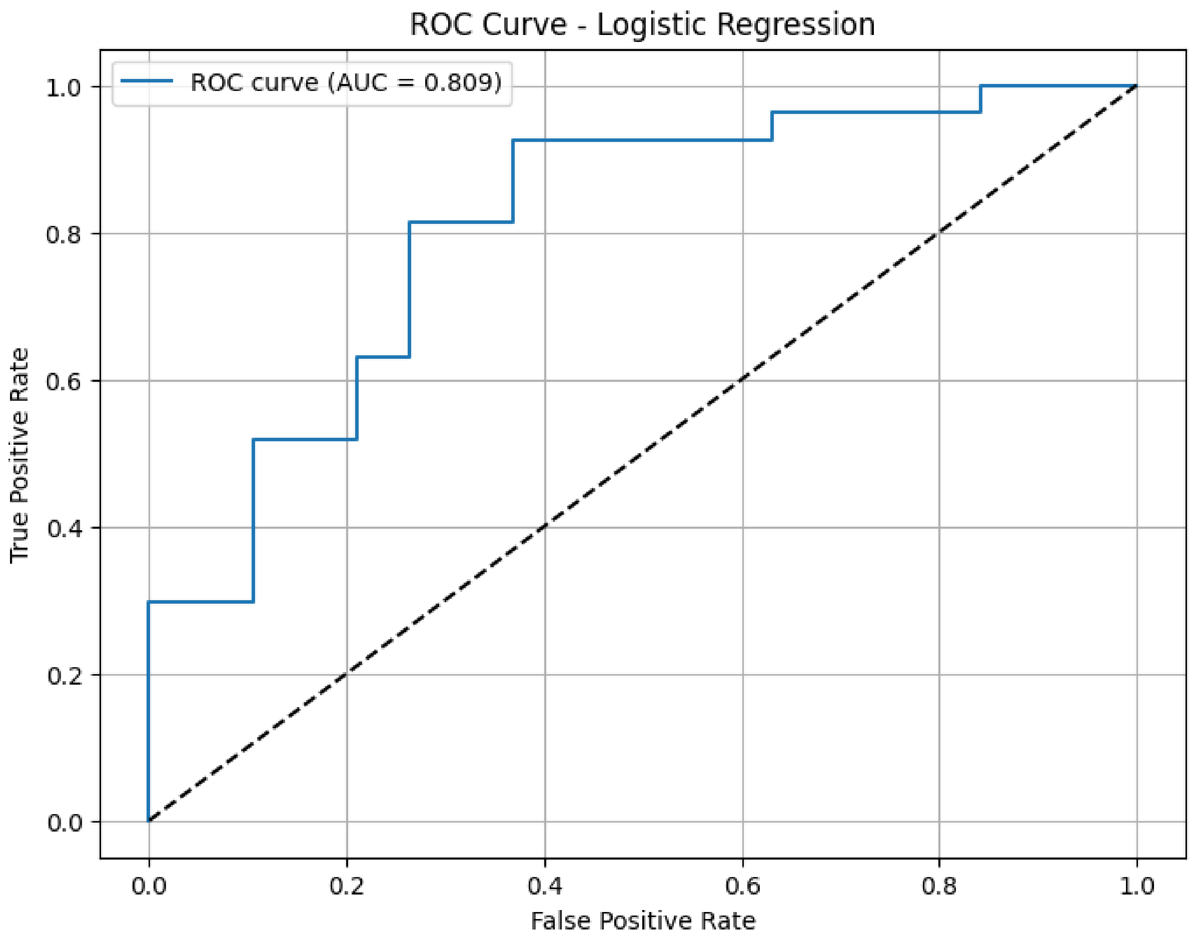


Figure 4.7: ROC curve of Frequentist Logistic Regression Model

Source: Own processing in Python

The graph represents the ROC curve (Receiver Operating Characteristic) for logistic regression. The AUC (Area Under Curve) is 0.809, indicating good model performance. The False Positive Rate (FPR) is plotted on the X-axis, and the True Positive Rate (TPR) is plotted on the Y-axis. The dashed diagonal line represents a random classifier (AUC = 0.5).The curve has a stepwise shape, which is typical for small datasets. Its position well above the diagonal line indicates the model's strong ability to distinguish between classes. A steep rise at the beginning of the curve, around FPR = 0.2, indicates that the model is able to correctly classify many positive instances with a low level of false positives.

In this chapter, the classic application of logistic regression was examined, which achieved satisfactory results. However, we have not yet utilized all technically available information, as the research design assumes the existence of a similar prior study. The main objective of this work was to explore the potential of Bayesian statistics for incorporating prior information about the phenomenon into the model. Therefore, the following section will focus on Bayesian logistic regression.

## 4.3 Bayesian Logistic Regression with Non-informative Priors

The Bayesian approach to logistic regression differs from the classical frequentist approach in that it treats model parameters (e.g., coefficients) as random variables with a certain probability distribution. In Bayesian regression, a prior distributions of parameters are initially set to reflect prior knowledge or assumptions about the values of these parameters. This representation is then updated through the posterior distribution based on the data. This approach allows for uncertainty in the data and model structure, which is particularly useful when there are few observations or when the model may be sensitive to some of the data.

As a result of the Bayesian approach, instead of point estimates of the parameters, as in frequentist logistic regression, we obtain a posterior distribution for each parameter. This allows us not only to estimate the likely values of the parameters, but also to see how confidently the model estimates them. For example, a wide posterior interval may indicate high uncertainty in the parameter value(John K. Kruschke, 2010).

**Model definition** For the Bayesian logistic model, the same formula as in 9.1 will be used. However, the key difference is that non-informative priors will be specified beforehand, meaning those that do not convey any prior knowledge about the data. It is worth mentioning that at this stage, we will perform sampling to estimate the posterior distribution of the model parameters, as analytically calculating this distribution is often impossible due to the complexity of the integrals. The sampling will take place in the parameter space $\theta$ .For this purpose, so-called weakly informative priors were chosen for all parameters. In this case, a weakly informative prior is defined as $N(0, 10)$ , this assumes that the parameters are equally likely to be distributed, with their density corresponding to an almost straight line and $Uniform(0, 1)$ for binary features .

To evaluate the posterior distribution of the model parameters, 4 sampling chains will be used, each performing 1,200 iterations including 200 warm-up steps. The convergence of the chains will then be assessed using the Gelman-Rubin statistic ($\hat{R}$).

### 4.3.1 Regression Parameter Estimates of Bayesian Logistic Regression with Non-informative Priors

By performing Bayesian logistic regression on the main experimental dataset, we will receive the following table. If we compare it with Table 9.1, we can notice that the

| Parameter | Mean | Std | Median | 2.5% | 97.5% | $n_{\text{eff}}$ | $\hat{R}$ |
|---|---|---|---|---|---|---|---|
| Const | -2.66 | 0.46 | -2.64 | -3.56 | -1.79 | 1408.34 | 1.00 |
| Sex | 2.01 | 0.40 | 2.00 | 1.27 | 2.80 | 1551.70 | 1.00 |
| Cholesterol | -0.39 | 0.16 | -0.38 | -0.70 | -0.08 | 4059.05 | 1.00 |
| FastingBS | 1.34 | 0.36 | 1.33 | 0.63 | 2.06 | 4252.68 | 1.00 |
| ExerciseAngina | 1.78 | 0.32 | 1.78 | 1.13 | 2.37 | 3667.68 | 1.00 |
| ChestPainType_ATA | -1.58 | 0.41 | -1.57 | -2.37 | -0.75 | 3529.06 | 1.00 |
| ChestPainType_NAP | -2.05 | 0.35 | -2.04 | -2.72 | -1.37 | 2915.19 | 1.00 |
| ST_Slope_Flat | 2.86 | 0.32 | 2.86 | 2.24 | 3.48 | 2565.34 | 1.00 |

Table 4: Regression Parameter Estimates of Bayesian Logistic Regression with non-informative priors

Bayesian model includes more detailed information about the distribution of parameters, including medians, standard deviations, and confidence intervals. In the first model, the coefficients are estimated using p-values to determine the significance of features, whereas in the second model, significance is reflected through the confidence intervals for each weight. The table also includes Gelman-Rubin statistics, with the values located in the `r_hat` column. A value of 1 indicates that all chains are well-mixed and have converged.

### 4.3.2 Confusion Matrix

Analyzing the confusion matrix and comparing it with Figure 9.2, it can be observed that non-informative priors showed the same results as frequentist logistic regression. Such a result is expected, as non-informative priors do not introduce additional information into the model.

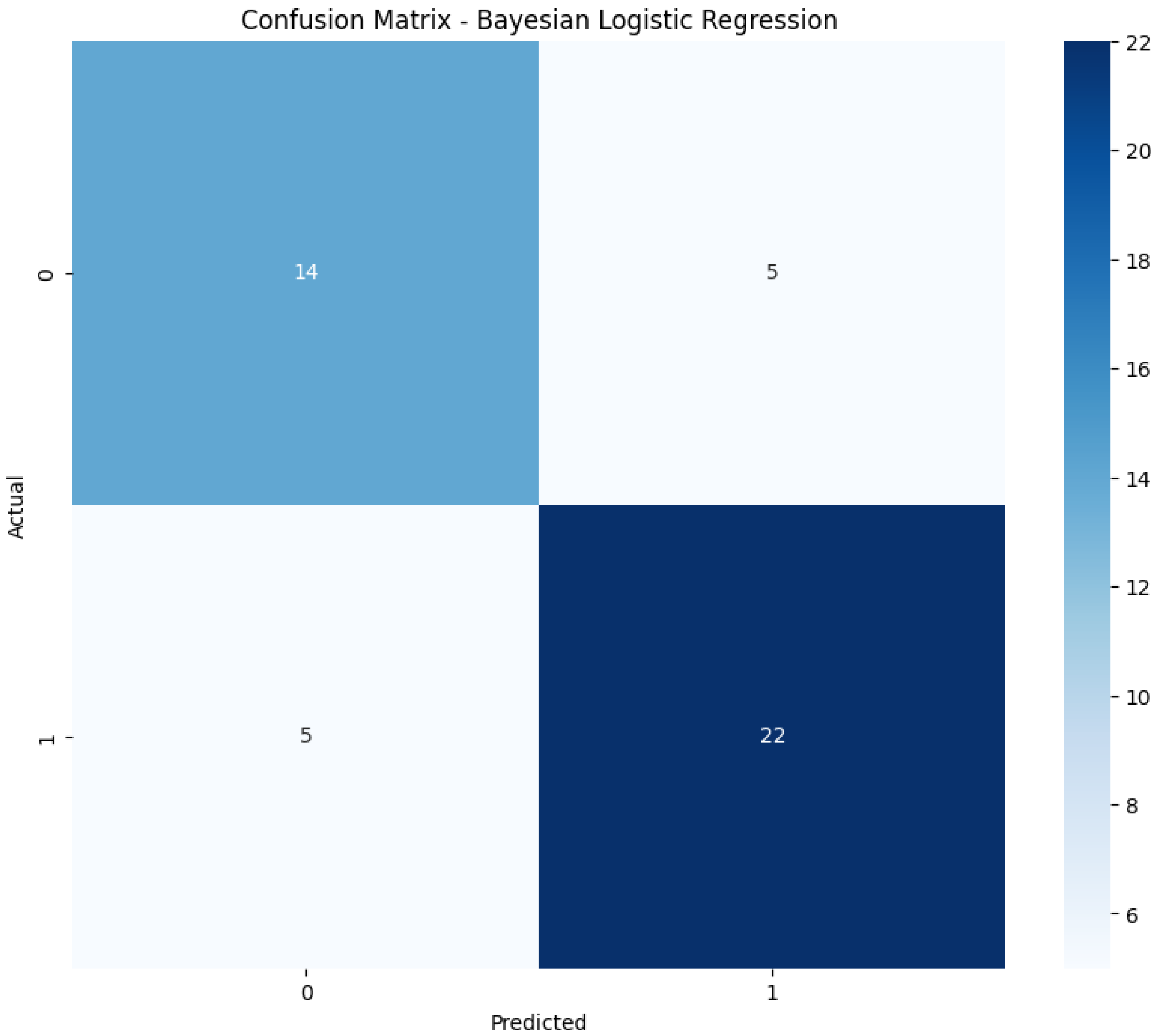


Figure 4.8: Confusion Matrix of Bayesian Logistic Regression Model with non-informative priors

Source: Own processing in Python

As can be seen from the confusion matrix, the model performs slightly worse in predicting the TN class (65 compared to 67 in the initial model) and is more focused on the TP class (95 compared to 90). Additionally, more patients are classified as FP (17 compared to 10), while fewer are classified as FN (7 compared to 10). This configuration is more favorable in the context of medical data analysis, as class 1 is more critical than class 0.

### 4.3.3 Metrics of the Model

The following table presents the metrics of the given model.

Table 5: Model Performance Metrics of Bayesian Logistic Regression with Non-informative Priors

| **Metric** | **Value** |
|---|---|
| Accuracy | 0.78 |
| Precision | 0.81 |
| Recall | 0.81 |
| F1-score | 0.81 |
| ROC AUC | 0.81 |

Source: Own processing in Python

Values of metrics of this model are the same as for frequentist logistic regression, which is expected since non-informative priors contribute no additional information to the model and thus asymptotically converge to maximum likelihood estimates under the frequentist approach.

### 4.3.4 ROC-curve

In the context of model metric analysis, it would also be interesting to analyze the ROC curve:

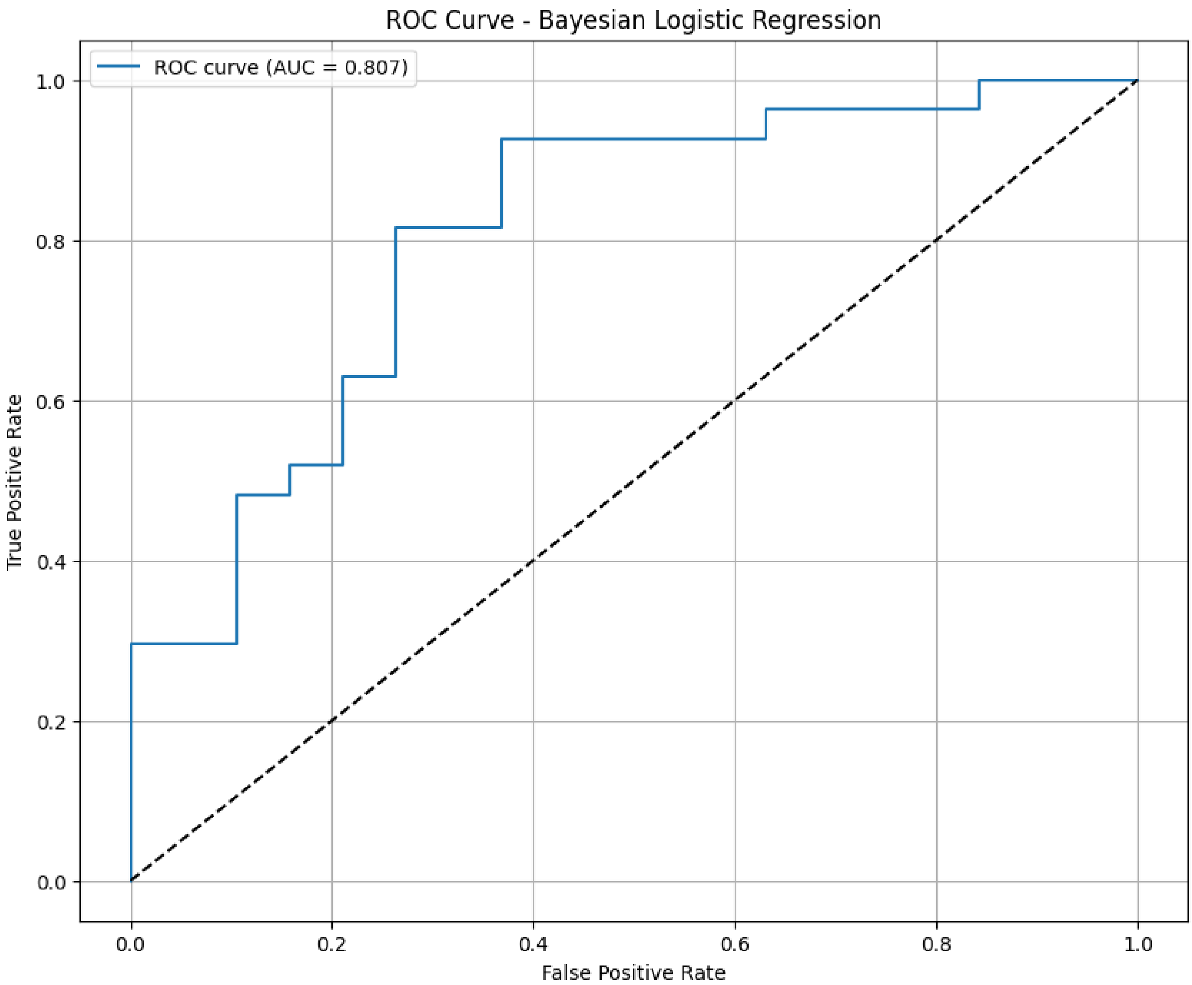


Figure 4.9: ROC curve of Bayesian Logistic Regression Model with Non-informative Priors

Source: Own processing in Python

The presented ROC curve for Bayesian logistic regression with non-informative priors exhibits nearly identical characteristics to the classical version of the model. The AUC value is 0.807, which is very close to the classical model's AUC of 0.809.

This similarity in results is not accidental and confirms the well-known fact that when non-informative priors are used, Bayesian logistic regression tends to yield parameter estimates similar to those obtained by the maximum likelihood method used in the classical version. This observation aligns with theoretical expectations, as in the absence of prior information (non-informative priors), the Bayesian approach relies solely on the data, leading to results similar to those of the classical approach.

The shape of the curve, including all key inflection points and the overall growth pattern, almost entirely mirrors the graph of the classical model, visually confirming the equivalence of the two approaches in this case.

### 4.3.5 Trace Plots and Kernel Density Estimation

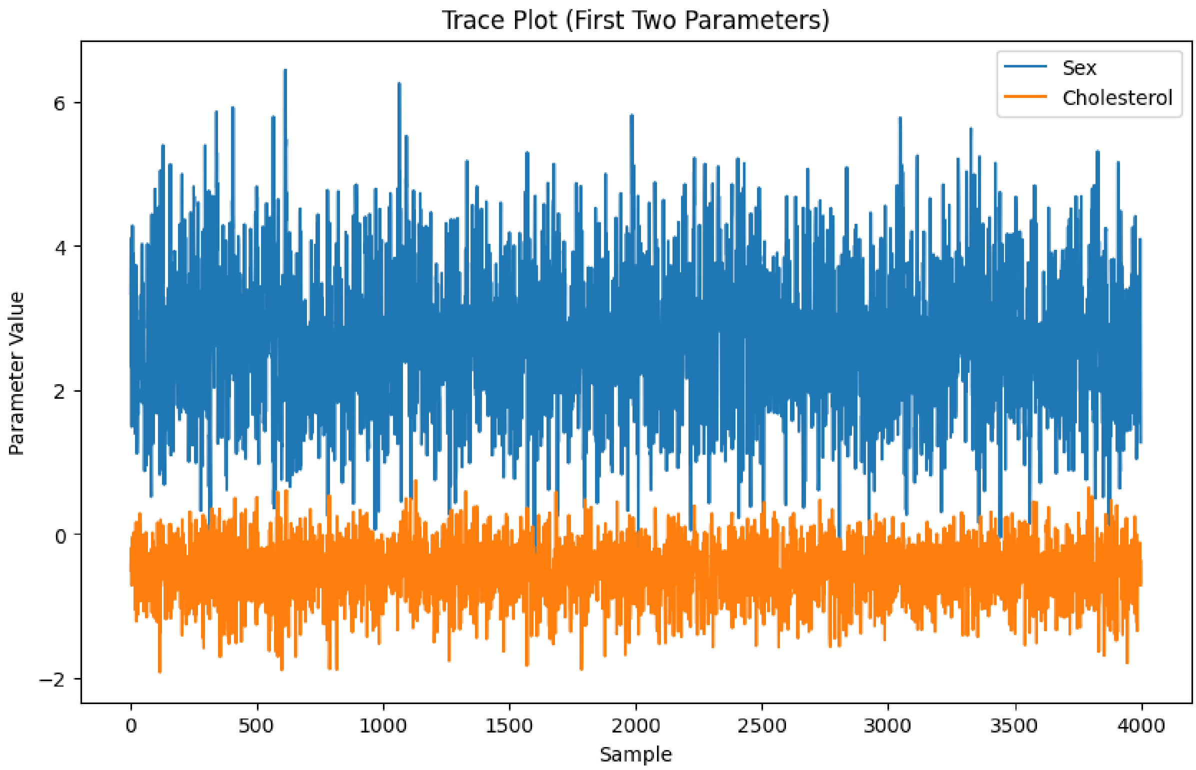


Figure 4.10: Trace Plot and Kernel Density Estimation for Coefficient Sex and Cholesterol

Source: Own processing in Python

The presented graph shows the results of MCMC sampling for two parameters within the framework of Bayesian analysis. This graph, known as a Trace Plot, illustrates the dynamics of parameter values for Sex and Cholesterol over 4,000 iterations. The Sex parameter (blue line) fluctuates within a range from 1 to 6, while the Cholesterol parameter (orange line) varies between -2 and 1. Both chains exhibit stable behavior without noticeable trends. The quality of sampling is assessed as good due to the even "mixing" of values, the absence of a required burn-in period, and the uniform distribution of points without pronounced clusters, gaps, or anomalous outliers. Such behavior of the Markov chains indicates successful algorithm convergence and the reliability of the obtained results. The stationarity of the process and the stability of fluctuations around the mean values confirm that the sample thoroughly explores the parameter space of the model.

### 4.3.6 Posterior Distributions

The following graphs represent the posterior distributions of model parameters obtained through Bayesian analysis. Each graph shows the density of the distribution (blue shading), the mean value of the parameter (red vertical line), the 95% credible interval (dashed red lines), and the zero value (dashed gray line).

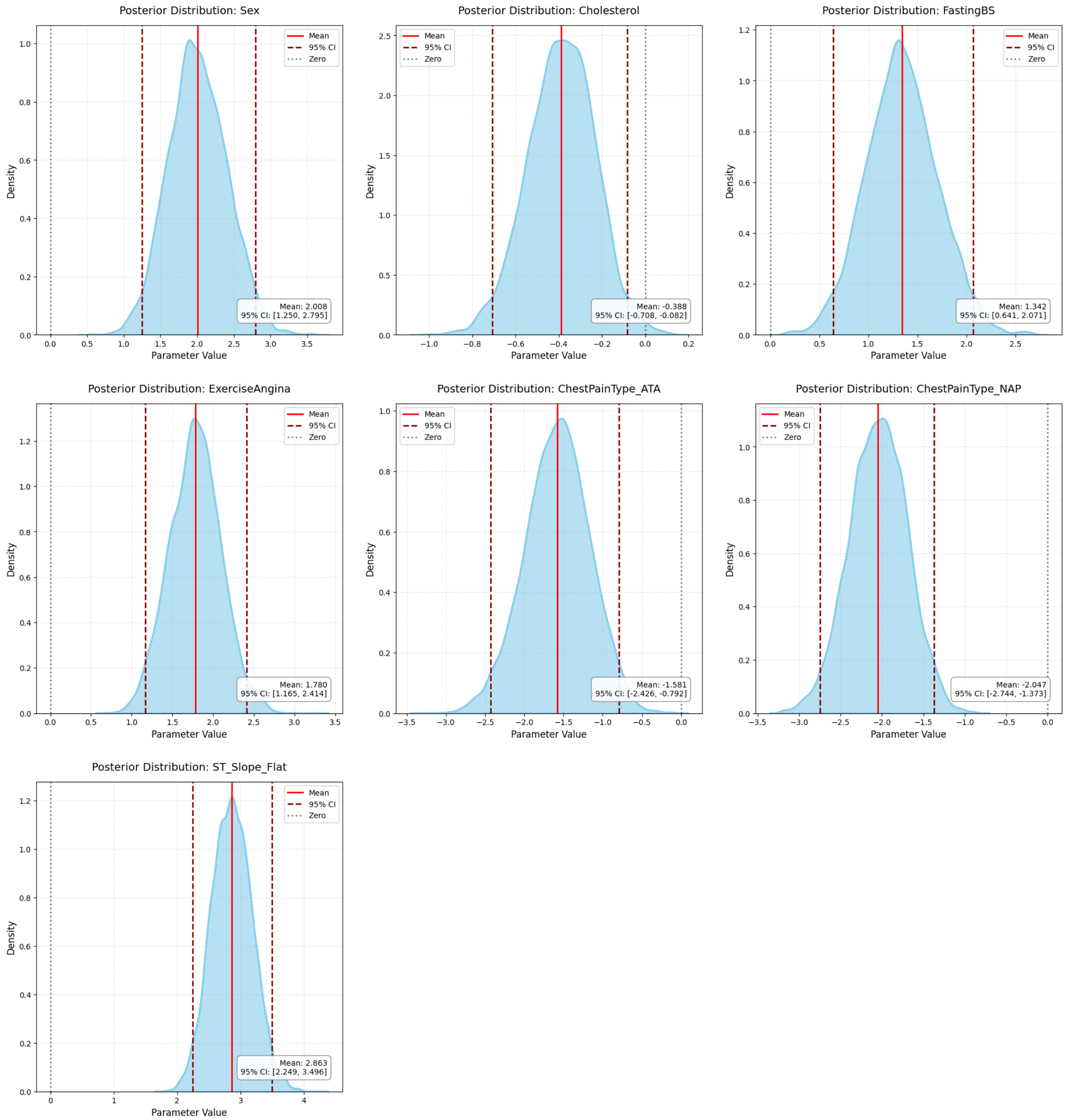


Figure 4.11: Posterior Distribution Graphs of Bayesian Logistic Regression with Non-informative Priors

Source: Own processing in Python

The distributions for the parameters Sex and ExerciseAngina exhibit symmetric shapes with pronounced peaks. For the parameter Cholesterol, the mean value is approximately -0.388, with a credible interval ranging from -0.708 to -0.082. The parameters ChestPainType_ATA and ChestPainType_NAP have distinctly negative values, indicating a negative correlation with the target variable. The distribution of FastingBS is shifted towards the positive region, with a mean value exceeding 1. The distribution of the parameter ST_Slope_Flat is narrower, reflecting greater certainty in its estimate.

Overall, the distributions appear unimodal and smooth, indicating good convergence of the MCMC algorithm. The credible intervals reflect the level of uncertainty in parameter estimates, and their positions relative to zero help assess the statistical significance of each parameter's influence.

## 4.4 Bayesian Logistic Regression with Informative Priors

After applying logistic regression to the large dataset, we obtained external information about the model parameters. Based on these Bayesian logistic regression results, we extracted posterior distribution statistics, specifically the mean values and standard deviations, to use as informative priors in the new model. The priors were specified as $\mathcal{N}(\mu_{\text{previous}}, \sigma_{\text{previous}})$.

### 4.4.1 Regression Parameter Estimates of Bayesian Logistic Regression with Informative Priors

The resulting coefficient table is presented below.

| Parameter | Mean | Std | Median | 2.5% | 97.5% | $n_{\text{eff}}$ | $\hat{R}$ |
|---|---|---|---|---|---|---|---|
| Const | -2.91 | 0.32 | -2.91 | -3.50 | -2.25 | 3666.30 | 1.00 |
| Sex | 1.69 | 0.31 | 1.69 | 1.08 | 2.30 | 3948.45 | 1.00 |
| Cholesterol | -0.50 | 0.14 | -0.50 | -0.77 | -0.24 | 7164.36 | 1.00 |
| FastingBS | 1.40 | 0.32 | 1.39 | 0.76 | 2.03 | 6506.53 | 1.00 |
| ExerciseAngina | 1.58 | 0.27 | 1.58 | 1.03 | 2.09 | 5660.41 | 1.00 |
| ChestPainType_ATA | -1.67 | 0.36 | -1.67 | -2.37 | -0.94 | 6379.42 | 1.00 |
| ChestPainType_NAP | -1.59 | 0.31 | -1.59 | -2.20 | -0.98 | 5400.88 | 1.00 |
| ST_Slope_Flat | 2.81 | 0.28 | 2.82 | 2.26 | 3.32 | 6550.63 | 1.00 |

Table 6: Regression Parameter Estimates of Bayesian Logistic Regression with iInformative Priors

Source: Own processing in Python

The analysis of the Bayesian logistic regression results shows that all model parameters are statistically significant, as their 95% confidence intervals do not include zero. The parameter ST_Slope_Flat has the greatest influence on the target variable, with a coefficient of 2.81, while Cholesterol and both types of ChestPainType exhibit a negative impact on the outcome.

The parameter estimates are characterized by good accuracy, as indicated by relatively small standard deviations. The least uncertainty is observed for the Cholesterol parameter (0.14), while ChestPainType_ATA shows the highest uncertainty (0.36). The convergence quality of the MCMC algorithm is confirmed by Rˆ values close to 1.0 for all parameters, as well as large effective sample size (ESS) values.

In terms of interpretation, the constant (-2.91) reflects the baseline log-odds, and the positive coefficients for the parameters Sex, FastingBS, ExerciseAngina, and ST_Slope_Flat indicate an increased likelihood of a positive outcome with their increase. In contrast, the negative coefficients for Cholesterol and both types of ChestPainType suggest a decreased likelihood of a positive outcome as these variables increase. These results support the reliability of the model and provide a clear understanding of the influence of various predictors on the predicted variable.

### 4.4.2 Confusion Matrix

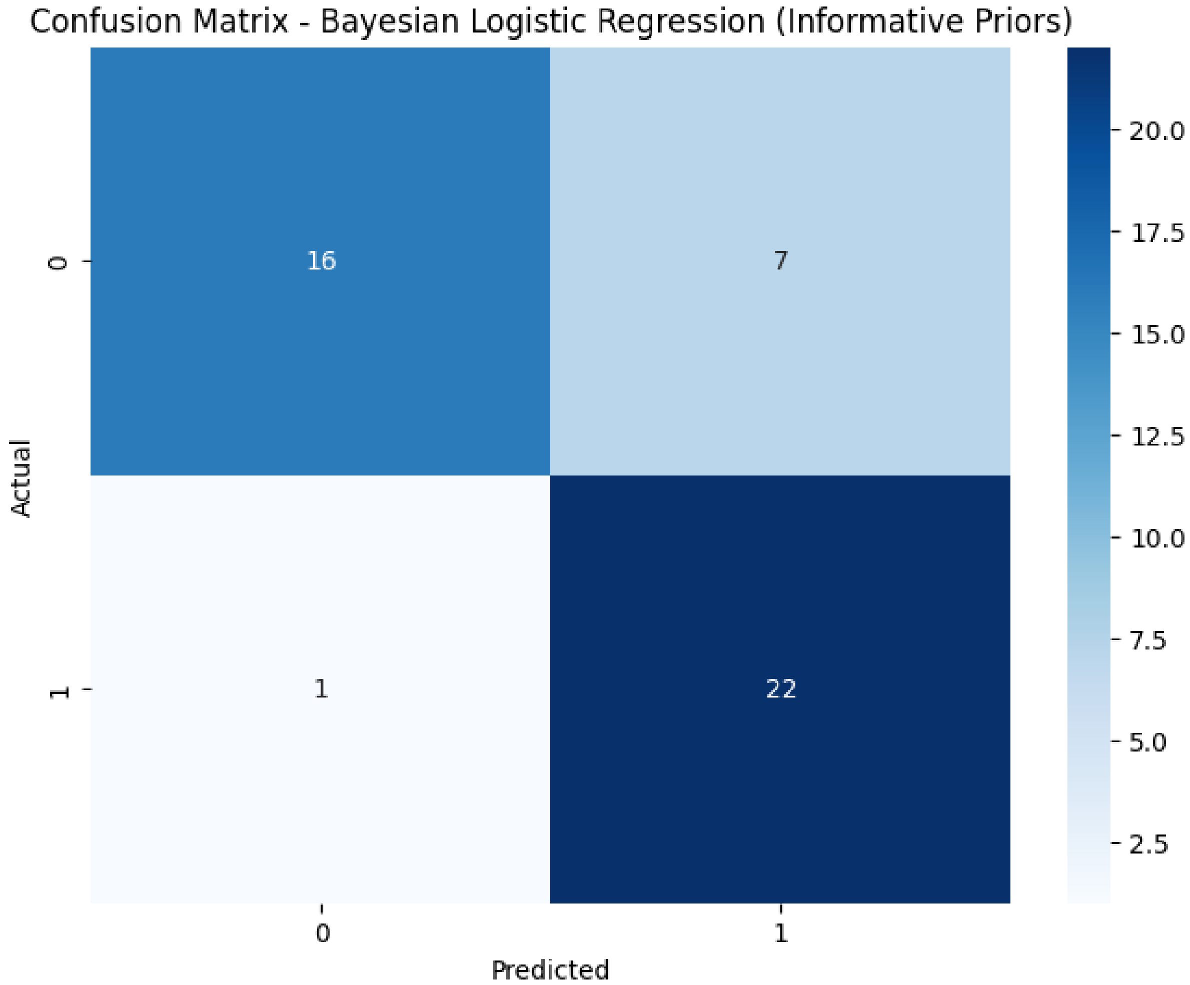


Figure 4.12: Confusion Matrix of Bayesian Logistic Regression Model with Informative Priors

Source: Own processing in Python

When comparing the two Bayesian logistic regression models, it is evident that the version with informative prior distributions yields more favorable results for our specific case with the HeartDisease dataset. It is particularly important to emphasize that in the context of diagnosing heart disease, false negatives (missing a true diagnosis) are significantly more critical than false positives.

The model with informative priors demonstrates a better balance in this regard: with the same number of correctly identified disease cases (22), it allows for only 1 false negative, compared to 5 in the baseline model. Although this results in a slightly higher number of false positives (7 against 5), this trade-off is fully justified from a clinical perspective, where missing a potential heart disease diagnosis can have serious consequences for the patient.

Thus, despite a slight increase in false positives, the informative model performs better in achieving the primary goal—minimizing the miss of true disease cases, making it more suitable for practical use in medical diagnostics.

### 4.4.3 Metrics of the Model

Priors that are informative with respect to the outcome improve predictive performace of the model on several metrics. The model achieves an accuracy of 0.83 which implies that 83% of the time all cases are correctly classified. The precision of 0.76 implies that

| Metric | Informative Priors |
|---|---|
| Accuracy | 0.83 |
| Precision | 0.76 |
| Recall | 0.96 |
| F1-score | 0.85 |
| ROC-AUC | 0.93 |

Table 7: Performance metrics with Informative Priors

Own Processing in Python

when the model classifies a case as positive (heart disease) there is a 76% chance that it is correct. A very high score for recall of 0.96 means that 96% of all true positives are correctly predicted by the model, which means there are very few false negatives. In terms of the alignment between precision and recall, the F1 score of 0.85 indicates that the balance is quite good. Acceptance that the ROC – AUC value of 0.93 seems good when interpreting its effectiveness, it is also correct that the model discriminates cases from non-cases quite well.

#### 4.4.4 ROC-curve

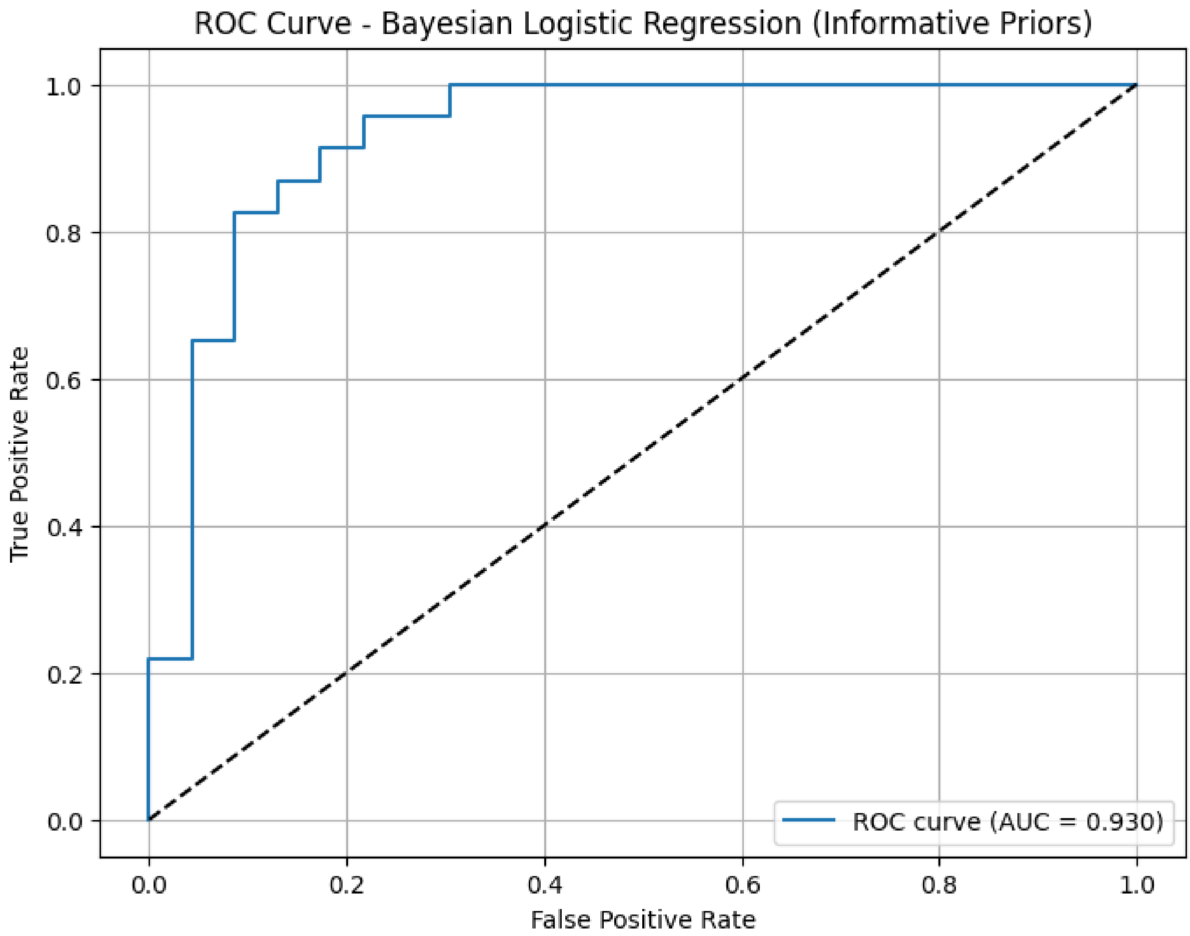


Figure 4.13: ROC Curve for Bayesian Logistic Regression With Informative Priors

Source: Own processing in Python

The presented graph shows the ROC curve for the Bayesian logistic regression model with informative prior distributions.Similarly, this graph reflects the characteristics of the

Bayesian logistic regression model. As in the previous case, a high AUC value (0.930) is observed, confirming the stability and reliability of the chosen approach. The shape of the curve retains the typical features of a high-quality classifier—noticeable performance above the diagonal line of random choice and a sharp rise in the initial part.The characteristic step-like curve is also preserved, reflecting the relatively small number of variables.

### 4.4.5 Trace Plots and Kernel Density Estimation

In conclusion of this chapter, we will perform an analysis of Trace Plots and Kernel Density Estimation.

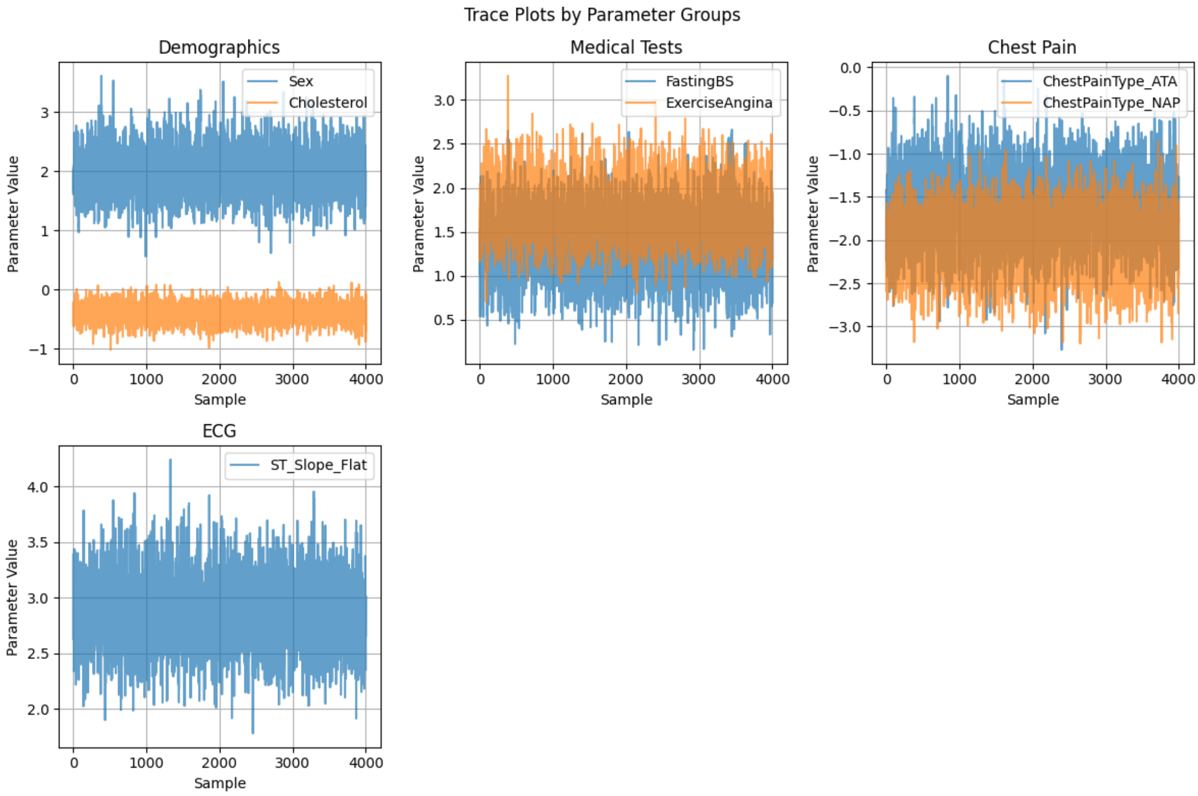


Figure 4.14: Trace Plots and Kernel Density Estimation of Parameters

Source: Own processing in Python

The presented set of Trace Plots shows the behavior of various parameters of the Bayesian model over time. The graphs indicate stable behavior across all parameters, reflecting good model convergence. Each parameter oscillates around its mean value: the parameter Sex stabilizes around 2, Cholesterol is centered near -0.5, while FastingBS and ExerciseAngina exhibit steady oscillations in the ranges of 0.5–1.5 and 1.5–2.5, respectively. Parameters associated with chest pain types (ChestPainType ATA and NAP) show consistent fluctuations in the negative range, and ST_Slope_Flat varies within the range of 2.0 to 4.0.

The frequent oscillations of parameter values indicate good "mixing," meaning the model is effectively exploring the parameter space. Low visual autocorrelation between successive iterations is another positive sign of model performance. Grouping parameters reveals that demographic factors exhibit varying scales of influence, medical test results display similar oscillation patterns, and chest pain-related parameters behave consistently. Notably, the ECG parameter shows the largest amplitude of fluctuations among all metrics.

These observations confirm the reliability of parameter estimates and the overall proper functioning of the model.

### 4.4.6 Posterior Distributions

The following graph depicts the posterior distributions of the parameters.

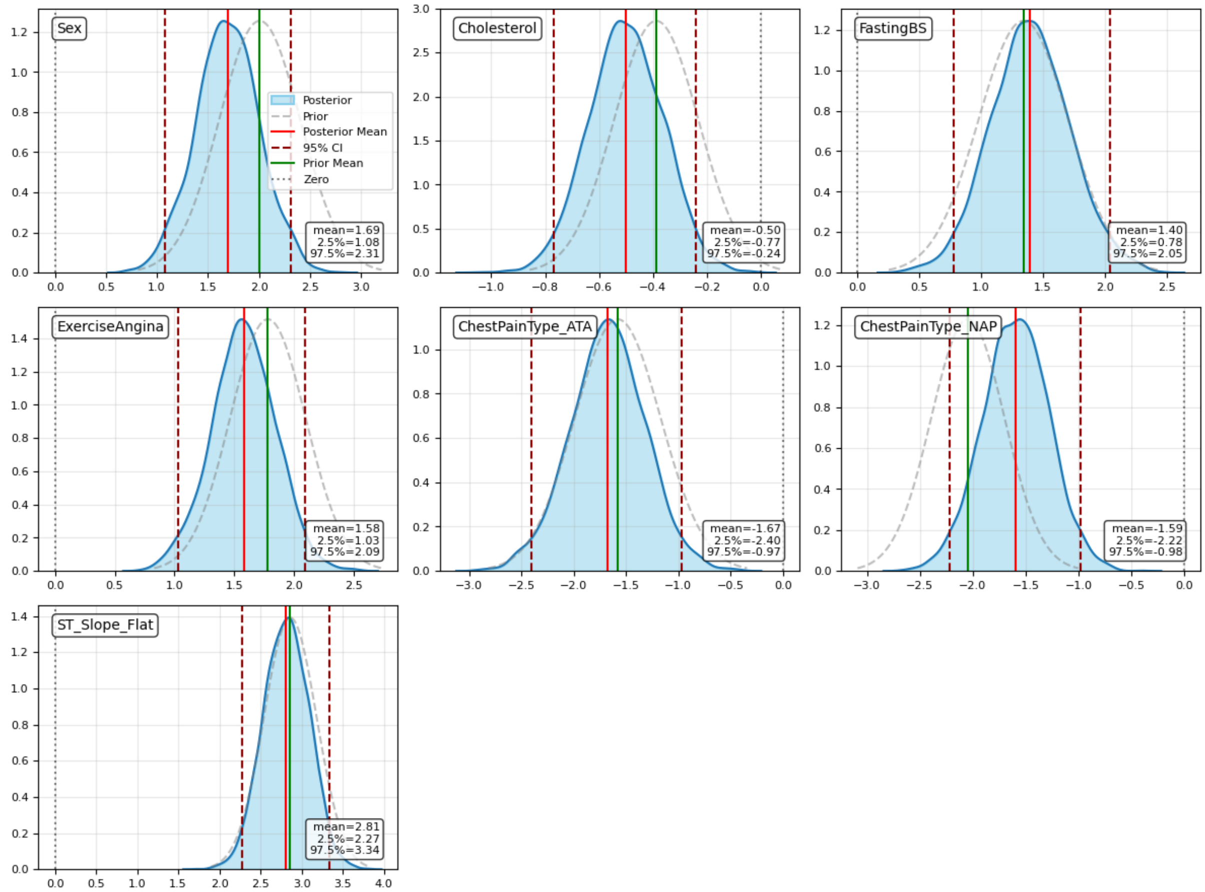


Figure 4.15: Posterior Distributions of Bayesian Logistic Regression Model with Informative Priors

Source: Own processing in Python

The analysis of the graphs reveals significant changes between prior and posterior distributions for all examined variables. Overall, there is a substantial reduction in uncertainty, as posterior distributions are narrower and more concentrated around their mean values compared to prior distributions.

For the variable Sex, the posterior distribution is centered around a mean of 1.69 with a noticeably narrower 95% confidence interval (1.08, 2.31). Cholesterol shows a leftward shift in its distribution with a mean value of -0.50, indicating a negative correlation. FastingBS exhibits a more defined distribution shape with a mean of 1.40 and significantly reduced variance. ExerciseAngina achieves a mean of 1.58 with a clearer distribution structure. The variables ChestPainType_ATA and ChestPainType_NAP show negative mean values (-1.67 and -1.59, respectively) with a noticeable narrowing of confidence intervals.

Notably, ST_Slope_Flat stands out, displaying the narrowest posterior distribution among all variables with a mean value of 2.81.

These observations highlight the impact of Bayesian inference in reducing uncertainty and refining parameter estimates, particularly by leveraging the data to adjust the initial priors. This reduction in uncertainty is crucial for improving the interpretability and reliability of the model.

# 5 Comparison of models

## 5.1 Metrics

The following table contain described metrics of the three Bayesian logistic regression models.

| **Metric** | **Frequentist Logistic Regression** | **Non-informative Priors** | **Informative Priors** |
|---|---|---|---|
| Accuracy | 0.78 | 0.78 | 0.83 |
| Precision | 0.81 | 0.81 | 0.76 |
| Recall | 0.81 | 0.81 | 0.96 |
| F1-score | 0.81 | 0.81 | 0.85 |
| ROC-AUC | 0.81 | 0.81 | 0.93 |

Table 8: Comparison of metrics between Frequentist and Bayesian logistic regression models with different priors.

Source: Own processing in Python

The analysis of our logistic regression models revealed some interesting findings. When comparing the traditional frequentist approach with two Bayesian variants (using non-informative and informative priors), we observed several key patterns.

First, it's worth noting that the frequentist model and the Bayesian model with non-informative priors performed exactly the same across all metrics. This isn't surprising, as non-informative priors essentially mimic the frequentist approach by not incorporating any prior knowledge into the model.

However, things got interesting when we looked at the model with informative priors. This version showed notable improvements in most of our performance metrics. The accuracy jumped from 78% to 83%, and we saw a dramatic increase in recall, which went from 81% all the way up to 96%. The ROC-AUC score also improved significantly, rising from 0.81 to 0.93, indicating better overall discriminative ability.

The only slight downside was a small drop in precision, which decreased from 81% to 76%. However, this trade-off seems acceptable given the substantial gains in other areas, particularly since the F1-score (which balances precision and recall) still improved from 0.81 to 0.85.

What makes these results particularly meaningful is that they validate our approach of incorporating prior knowledge into the model. The significant improvements in recall and ROC-AUC suggest that our prior information was indeed valuable and helped the model make better predictions, especially in identifying positive cases.

Based on these results, I would recommend using the model with informative priors, particularly if missing positive cases (false negatives) is more costly than having false positives. The substantial improvement in classification ability, as shown by the ROC-AUC score, makes this model a more robust choice for real-world applications where balanced performance is crucial. This results demonstrate the potential benefits of incorporating domain knowledge through informative priors in Bayesian modeling, especially when that prior information is well-founded and relevant to the problem.

## 5.2 ROC-curve

Figure 7.1 illustrates the Receiver Operating Characteristic (ROC) curve computed on the training dataset for all three models.

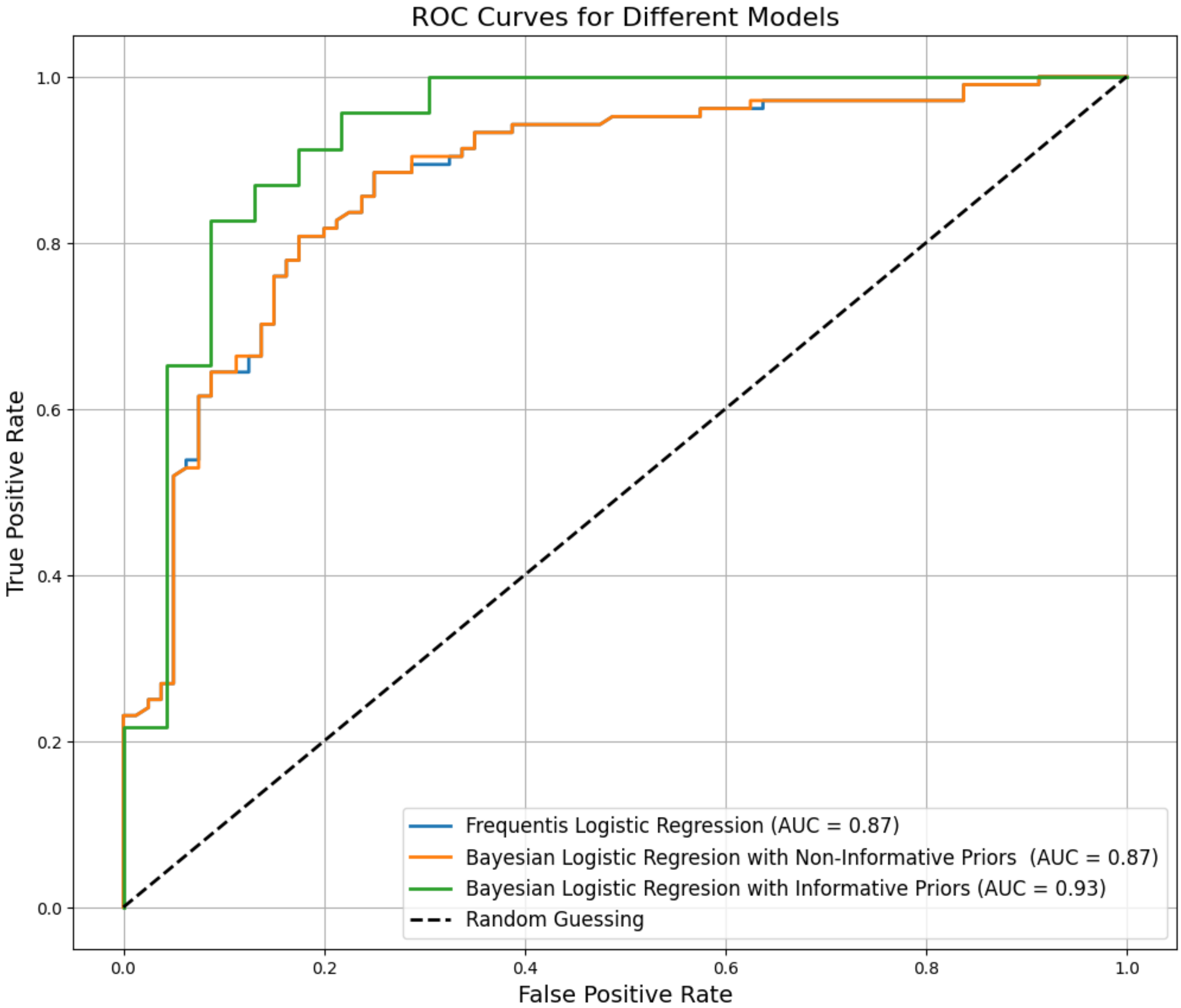


Figure 5.1: ROC Curve of Bayesian Logistic Regression Model with informative priors on Test data

Source: Own processing in Python

An analysis of the chart shows that the model with informative priors performs the best, achieving an AUC value of 0.93. The frequentist logistic regression and the Bayesian model with non-informative priors show nearly identical results with an AUC of 0.87. The reason behind this fact was described earlier. The advantage of the model with informative priors is especially noticeable on the left side of the graph, where, at a low False Positive Rate, it achieves a higher True Positive Rate. For example, at a False Positive Rate of about 0.2, the model reaches a True Positive Rate of approximately 0.9.

## 5.3 Comparison of Interval Estimates

In this chapter, we focus on the analysis of confidence interval (CI) lengths, which allows us to assess the level of certainty in the parameter estimates. The narrower the CI, the greater the confidence in the estimated value of the parameter, and vice versa.

When comparing the three approaches - classical logistic regression analysis, Bayesian approach with non-informative priors, and Bayesian approach with informative priors - we observe interesting dynamics in how confidence and credible interval lengths change.The

| Variable | CI Frequentist | Credible Interval Non-informative | Credible Interval Informative |
|---|---|---|---|
| const | [-3.453, -1.712]<br>1.74 | [-3.56, -1.79]<br>1.77 | [-3.50, -2.25]<br>1.25 |
| Sex | [1.185, 2.716]<br>1.53 | [1.7, 2.8]<br>1.53 | [1.08, 2.30]<br>1.22 |
| Cholesterol | [-0.368, 0.070]<br>0.60 | [-0.70, 0.08]<br>0.62 | [-0.77, -0.24]<br>0.53 |
| Fasting_BS | [0.583, 2.018]<br>1.43 | [-0.63, 2.706]<br>1.43 | [0.76, 2.03]<br>1.27 |
| ExerciseAngina | [1.123, 2.341]<br>1.22 | [1.13, 2.37]<br>1.24 | [1.03, 2.09]<br>1.06 |
| ChestPainType_ATA | [-2.353, -0.734]<br>1.62 | [-2.37, -0.75]<br>1.62 | [-2.37, -0.94]<br>1.43 |
| ChestPainType_NAP | [-2.687, -1.323]<br>1.36 | [-2.72, -1.37]<br>1.35 | [-2.20, -0.98]<br>1.22 |
| ST_Slope_Flat | [2.177, 3.420]<br>1.24 | [2.24, 3.48]<br>1.24 | [2.26, 3.32]<br>1.06 |

Table 9: Comparison of Confidence and Credible Intervals $\alpha = 0.05$ across Three Methods with their Lengths

Source: Own processing in Python

most notable improvement in estimation precision is observed for the model constant. When transitioning to non-informative priors, the credible interval length slightly increases by 1.7% (from 1.74 to 1.77), however, using informative priors leads to a substantial narrowing of the interval by 28.2% (to 1.25) compared to coinfidence interval calculated by the classical approach.

For the Sex variable, we see a stable interval length of 1.53 when using both the classical approach and non-informative priors. However, applying informative priors reduces the credible interval length by 20.3% (to 1.22), significantly improving the precision of estimating gender's influence on disease probability. Cholesterol shows the narrowest intervals among all variables. There is a slight increase in interval length by 3.3% (from 0.60 to 0.62) when using non-informative priors, but subsequent application of informative priors leads to a reduction of 11.7% (to 0.53) compared to the classical approach.

Fasting blood sugar (Fasting_BS) and chest pain type (ChestPainType_ATA) show similar dynamics: the interval length remains unchanged when transitioning to non-informative priors (1.43 and 1.62 respectively) but decreases by 11.2% and 11.7% when using informative priors. For exercise angina (ExerciseAngina), we observe a small increase in interval length by 1.6% (from 1.22 to 1.24) when using non-informative priors, however, informative priors allow for a reduction of 13.1% (to 1.06) compared to the classical approach.

Of particular interest is the ST_Slope_Flat variable, where the interval length remains unchanged (1.24) when using non-informative priors but decreases by 14.5% (to 1.06) when applying informative priors.

On average, the transition from the classical approach to Bayesian with informative priors leads to a reduction in confidence interval lengths by 15.1%. Meanwhile, using non-informative priors yields minimal changes (average increase of 0.7%). This suggests that the main gain in precision is achieved through the use of informative priors rather than the Bayesian approach itself. The greatest improvement in precision is observed for the model constant (28.2%) and the Sex variable (20.3%), while the smallest improvement is shown by the ChestPainType_NAP variable (10.3%). Such substantial improvement in estimation precision when using informative priors can have important practical implications for clinical interpretation of results and decision-making based on them.

# Conclusion

This research has provided a comprehensive comparison between frequentist and Bayesian approaches to logistic regression, with particular emphasis on their application to medical data. The results demonstrate several key findings that contribute to both theoretical understanding and practical applications in the field.

Our analysis reveals that Bayesian logistic regression with informative priors demonstrates superior performance compared to both classical logistic regression and Bayesian approaches with non-informative priors, particularly in scenarios with limited data. This is evidenced by improved performance metrics, with the informative prior model achieving higher accuracy (0.83 versus 0.78) and notably better ROC-AUC scores (0.93 versus 0.81).

One of the most significant findings is the substantial reduction in parameter uncertainty, as demonstrated by the narrowing of confidence intervals. The notable examples are observed in the Sex and ST_Slope_Flat variables, where the confidence interval length decreased by 20.3% (from 1.53 to 1.22) and 14.5% (from 1.24 to 1.06) when using informative priors respectively. This reduction in uncertainty has crucial implications for medical decision-making, where precise parameter estimates are essential for accurate risk assessment.

Based on these findings, several practical recommendations could be made:

- For scenarios with limited data availability and existence of external information, Bayesian logistic regression with informative priors should be considered as a primary approach, as it effectively leverages prior knowledge to compensate for data limitations
- The choice and calibration of prior distributions of parameters should be carefully considered, as it significantly impacts model performance and uncertainty estimates especially when we have a limited amount of data. .
- When working with medical data, the improved precision offered by Bayesian methods can be particularly valuable for clinical decision-making
- The computational cost of MCMC methods should be taken into account during project planning, as these methods are significantly more time-consuming compared to classical approaches. Even though modern computers are capable of handling complex computations efficiently, certain scenarios may still require additional resources, such as cloud computing, to manage the computational load. While this is not a significant technical constraint given the availability of advanced infrastructure today, relying on external computing solutions can increase costs and add logistical considerations, such as software integration and data security.

It is important to note that while Bayesian methods offer superior statistical properties, they come with increased computational overhead. The implementation of MCMC sampling methods, while necessary for posterior estimation, is computationally intensive and time-consuming. This trade-off between computational cost and statistical accuracy should be carefully considered in time-sensitive applications.

However, this research also opens up several promising directions for future investigation:

- Further exploration of prior specification, particularly examining how different standard deviations in prior distributions reflect varying degrees of confidence in historical data

- Investigation of how the relationship between sample size and prior information affects model performance
- Development of systematic approaches for translating expert knowledge into informative priors
- Research into more computationally efficient MCMC algorithms and sampling methods to reduce execution time while maintaining statistical accuracy

The limitations of this study primarily relate to the specific context of medical data and computational constraints. While MCMC methods provide robust results, their time-consuming nature may limit their applicability in scenarios requiring rapid model updates or real-time predictions. Future work could explore the generalizability of these findings to other domains and investigate more efficient computational approaches.

In conclusion, this research project fulfilled its main goal by conducting a comparative analysis of frequentist and Bayesian approaches to logistic regression, applied to real-world medical datasets to evaluate the performance of each method. It highlighted the advantages of incorporating informative priors in Bayesian logistic regression, showcasing their potential to enhance model interpretability and improve predictive accuracy when prior knowledge is available. In conclusion, this study demonstrates the practical advantages of Bayesian logistic regression, particularly in scenarios with limited data availability. The ability to incorporate prior knowledge through informative priors not only improves model performance but also provides more precise parameter estimates, which is crucial in medical applications. While the computational overhead of MCMC methods presents a practical challenge, the improved statistical properties often justify the additional computational investment, especially in critical applications like medical diagnosis where accuracy is paramount. These findings contribute to the growing body of evidence supporting the practical utility of Bayesian methods in real-world applications, while also highlighting promising directions for future research in prior specification, uncertainty quantification, and computational efficiency.

# References


[1] Kolmogorov,A.N. (1933) Foundations of the Theory of Probability,Springer, Berlin, 1950. Chelsea Publishing , New York.

[2] Bolstad, W. M. (2007). Introduction to Bayesian statistics (2nd ed.). Hoboken: John Wiley.

[3] Bílková, D. (2009). Pravděpodobnost a statistika. Plzeň: Vydavatelství a nakladatelství Aleš Čeněk.

[4] Gravner, J. (2024, August 31). Lecture Notes for Introductory Probability. Mathematics Department, University of California, Davis.

[5] Johnson, A. A., Ott, M. Q., Dogucu, M. (2021). Bayes rules! An introduction to applied Bayesian modeling. CRC Press.

[6] Bertsekas, D. P., Tsitsiklis, J. N. (2000). Introduction to probability: Lecture notes for Course 6.041-6.431. Massachusetts Institute of Technology.

[7] Feng,C.(2024). MCMC Interactive Demo.

[8] Hoffman, M. D., Gelman, A. (2014). The no-U-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. Journal of Machine Learning Research, 15, 1593–1623.

[9] Jurafsky, D., Martin, J. H. (2024). Speech and language processing (3rd ed., draft). [August 20, 2024 release].

[10] Elisa T. Lee, J. W. (2003). Statistical Methods for Survival Data Analysis.

[11] Malá, I. (2024). Statistické úsudky (Druhé vydání.). Praha: Professional Publishing.

[12] Koop, G. Bayesian Econometric Methods. New York: Cambridge University Press, 2007. ISBN 0-521-67173-6

[13] Google. (n.d.). Machine learning crash course. URL:https://blog.google/technology/developers/machine-learning-crash-course/

[14] Aldraimli, M., Soria, D., Parkinson, J., Thomas, E. L. (2020). Machine learning prediction of susceptibility to visceral fat associated diseases. Health and Technology, 10(4), 925–944.

[15] Hosmer, D. W., Lemeshow, S. (2000). Applied logistic regression (2nd ed.). Wiley.

[16] Hastie, T., Tibshirani, R., Friedman, J. (2009). The elements of statistical learning: Data mining, inference, and prediction (2nd ed.). Springer.

[17] Caflisch, R. E. (1998). Monte Carlo and quasi-Monte Carlo methods. Ada Numerica, pp. 1–49. Cambridge University Press.

[18] Karel, T. (2017). Predikce vývoje HDP pomocí bayesovských vícerozměrných autoregresních modelů [Doctoral dissertation, Vysoká škola ekonomická v Praze, Fakulta informatiky a statistiky].

[19] Neal, R. M. (2011). MCMC using Hamiltonian dynamics. In S. Brooks, A. Gelman, G. Jones, X.-L. Meng (Eds.), Handbook of Markov Chain Monte Carlo (Chap. 5). Chapman Hall/CRC Press.

[20] Jaiswal, S. (2024, January 4). Normalization in machine learning: A simple guide. DataCamp. Retrieved October 2, 2024, from https://www.datacamp.com/tutorial/normalization-in-machine-learning

[21] Vetrov, D. P., Kropotov, D. A. (2011, November 19). Bayesian methods in machine learning, 2011: Markov Chain Monte Carlo methods (MCMC) [Lecture notes]((Методы Монте Карло по схеме марковской цепи)). Moscow State University, Faculty of Computational Mathematics and Cybernetics.

[22] Bernoulli, J. (1713/2005). The Art of Conjecturing (E. D. Sylla, Trans.). Springer. (Original work published 1713)

[23] Hebák, P. 2012. Srovnání Klasické a bayesovské pravděpodobnosti statistiky. Acta Oeconomica Pragensia. 1, 2012.

# List of Figures

# List of Tables